Fair Division Under Inaccurate Preferences

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Abstract

The fair allocation of scarce resources is a central problem in mathematics, computer science, operations research, and economics. While much of the fair division literature assumes that individuals have underlying cardinal preferences, eliciting exact numerical values is often cognitively burdensome and prone to inaccuracies.

In this paper, we explore the broad landscape of fair division of indivisible items given inaccurate cardinal preferences, with a focus on minimizing envy. We model these inaccuracies as additive noise, and explore a variety several settings based on whether true preferences and inaccuracies are each stochastic or worst-case. For stochastic true preferences, we show that envy-free allocations can be computed with high probability for both stochastic and worst-case noise. For worst-case true preferences and bounded noise, we analyze the maximum envy achieved by the well-known Round-Robin algorithm. This bound is shown to be tight for deterministic algorithms and we prove applications of this bound using known results from statistical estimation. Finally, we consider a setting with worst-case preferences and noise, where the true preferences for each item are revealed upon its allocation. Here, we give an efficient online algorithm that guarantees logarithmic maximum envy with high probability, for sufficiently bounded noise. This result builds upon and generalizes a known result from algorithmic discrepancy to a setting with noisy input data.

1 Introduction

Fair division studies the fundamental problem of fairly allocating resources among agents with heterogeneous preferences. Mathematical formulations date back to Steinhaus [35] and Dubins and Spanier [23], though some of the field's core questions (and even key algorithms) can be traced back to the Bible and ancient Greece.

Most of the existing literature assumes that agents' preferences are represented by valuation functions, which are either reported directly to the algorithm, or accessed through exact value queries. While we have a deep understanding of when fairness is and is not attainable in these models, requiring participants to report *exact* numerical values is a severe practical limitation. Eliciting precise cardinal valuations is cognitively burdensome and prone to inaccuracies. Thus, a growing body of work studies algorithms that only require *ordinal information*. This idea originates from the seminal work of Procaccia and Rosenschein [34], who introduced the concept of distortion: measuring the worst-case loss of a cardinal objective (e.g., utilitarian social welfare) due to relying solely on limited information, such as ordinal rankings.

Most research in this area takes a worst-case approach; an algorithm makes a decision based on ordinal information, and the underlying cardinal valuations are then chosen adversarially (subject to consistency with the reported rankings) to maximize the objective loss. While this approach yields

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robust guarantees, it is inherently pessimistic. Furthermore, purely ordinal models ignore the reality that participants often *can* provide useful, albeit imperfect, cardinal signals.

In this paper, we bridge the gap between the perfectly accurate cardinal and fully ordinal models, initiating the study of *fair division under inaccurate cardinal preferences*. Our objective is to design algorithms that minimize maximum envy (with respect to true preferences) using only noisy cardinal estimates as input. We explore several formalizations of this setting, capturing distinct noise models and assumptions about the agents' underlying true preferences.

Our central finding is that inaccurate cardinal information is remarkably powerful. Across a range of stochastic and worst-case models, pairing noisy cardinal estimates with carefully chosen algorithms achieves envy guarantees qualitatively comparable to those requiring exact cardinal access.

1.1 Our Contributions

We consider allocating m divisible items to n agents with additive preferences, and m, where we only receive a "noisy" estimate $\hat{v}_{i,j}$ of agent i's value for item j, instead of their "true" value $v^*_{i,j}$. Our goal is to allocate the items to minimize envy with respect to the true values $v^*_{i,j}$. We categorize our results based on whether the underlying true valuations are stochastic or worst-case.

Stochastic True Valuations. First, we explore scenarios where the true valuations $v^*_{i,j}$ are drawn from a distribution. We analyze two subsettings based on whether the additive noise is stochastic or worst-case. In both regimes, we establish that envy-free allocations exist with high probability.

First, we consider the case of stochastic true valuations and stochastic noise: Each agent has a true value for item j $v^*_{i,j} \sim \mathcal{D}$, and has noise $\eta_{i,j} \sim \mathcal{D}'$ generated, and we observe the noisy values $\hat{v}_{i,j} = v^*_{i,j} + \eta_{i,j}$; here $\mathcal{D}$, $\mathcal{D}'$ are two unknown, fixed, real-valued distributions. In Theorem 1, we show that simply maximizing social welfare with respect to these observed values yields an envy-free allocation with high probability. This generalizes a result by Dickerson et al. [22] to the setting with inaccurate preferences.[1] Our proof utilizes a strict statistical correlation inequality to show that agents in expectation prefer the items they receive to those received by other agents (even if noise mean its possible an item will go to an agent that has low "true" value for it). This, in turn, allows us to argue via concentration that agents strictly prefer their own bundle, even when the allocation decisions are based on the noisy values.

Next, we consider stochastic true valuations paired with worst-case noise. We work within a general framework where the value vector for each item j is drawn i.i.d. from a bounded, non-negative vector-valued distribution $\mathcal{D}$ over $[0, b]^n$. Here, agents' valuations for a single item may be arbitrarily correlated, but values across different items remain independent. We prove that if $\mathcal{D}$ satisfies a mild technical condition—essentially ensuring no two agents are identical up to scaling—and the noise is bounded by a constant, there exists an algorithm that outputs an envy-free allocation with high probability (Theorem 2). To prove this, we adapt a geometric existence result from the cake-cutting literature [13] to establish that strongly envy-free allocations exist with high probability in the underlying noiseless setting. Then, approximately minimizing envy even within the noisy values suffices to find an envy-free allocation.

Worst-Case True Valuations. In the second half of our results, we turn to scenarios where the underlying true valuations $v^*_{i,j}$ are fixed but unknown (i.e., worst-case).

We first pair worst-case true valuations with worst-case noise. Assuming the unobserved true values $v^*_{i,j} \in [0, b]$ are fixed, we receive estimates $\hat{v}_{i,j}$ satisfying $|\hat{v}_{i,j} - v^*_{i,j}| \leq \varepsilon$ for some (possibly unknown) noise bound $\varepsilon \geq 0$. Under this model, we derive a tight bound on the best possible envy guarantee achievable by any deterministic algorithm operating solely on the noisy estimates $\hat{v}_{i,j}$ (Theorems 3 and 4). Notably, we prove that the standard Round-Robin algorithm optimally matches this guarantee.[2]

We then present two applications of this Round-Robin optimality result.

In the first application (Application 1), we present an end-to-end framework where we use noisy pairwise comparisons to estimate values within an established framework [20, 21], and then use

[1] Dickerson et al. [22] show that envy-free allocations exist with high probability when $\eta_{i,j} = 0$ for all i, j.

[2] The Round-Robin algorithm proceeds in rounds: agents take turns in a fixed sequence $1, 2, \ldots, n$, with each agent selecting their most preferred available item based on the noisy estimates $\hat{v}_{i,j}$.

the estimated values to make allocation decisions. Specifically, we assume these stochastic ordinal observations are drawn via the Bradley-Terry-Luce (BTL) model (also known as the multinomial logit model)—arguably the most prevalent discrete choice model [19, 31–33, 37]. In the BTL model, the probability that an agent ranks one alternative above another is determined by applying the logistic function to the difference between their underlying cardinal valuations. Given such ordinal information, we first produce estimates $\hat{v}_{i,j}$ for the true preferences. Using this ordinal data, we generate accurate estimates $\hat{v}_{i,j}$ of the true preferences and feed them into the Round-Robin algorithm to produce an allocation with bounded maximum envy. One challenge is that BTL estimation inherently cannot detect constant shifts in an agents' values as it depends only on differences. A key property of Round-Robin is that it produces a *balanced* allocation,[3] which allows it to make strong envy guarantees even with this wrinkle.

The second application (Application 2) deals with stochastic noise with an *unbiased* error. Specifically, we assume that $\hat{v}_{i,j} := v^*_{i,j} + \varepsilon_{i,j}$, where $\varepsilon_{i,j} := \sigma_{i,j} \cdot z_{i,j}$ where $z_{i,j}$s are drawn i.i.d. from a non-negative (possibly unknown) distribution $\mathcal{D}$ and $\sigma_{i,j}$s are drawn uniformly at random from $\{-1, 1\}$. We consider small tail distributions $\mathcal{D}$, and in particular, MHR distributions (see Section 4.2 for the definition). We show that, given only noisy estimates, allocations with a bounded maximum envy can be computed efficiently. To prove this, we employ known results about the order statistics of MHR distributions and a recent result concerning the concentration of order statistics of MHR distributions. Combining this bound with our Round-Robin result yields the bounds.

Finally, we proceed to study a different, more permissive noise model. As before, we have to allocate items to agents based on the noisy estimates $\hat{v}_{i,j}$; however, this time, once any item j is allocated, then the true valuation vector v^*_j corresponding to this item is revealed.[4] This immediate feedback gives more power to the algorithm. In this model, we show that there is a simple, online algorithm that achieves a maximum envy bound of $O\left(\sqrt{n}(\log \frac{mn}{\delta} + m\varepsilon)\right)$ with probability at least $1 - \delta$ (Theorem 5), where the noise is bounded by ε, i.e., $|v^*_{i,j} - \hat{v}_{i,j}| \leq \varepsilon$. The algorithm achieving this bound is inspired by (and is a modified version of) an online discrepancy minimization algorithm given by Alweiss et al. [2].[5] Notably, if the noise satisfies the upper bound $\varepsilon \leq \frac{1}{m} \log \frac{mn}{\delta}$, then we get an envy as well as a discrepancy bound of $O\left(\sqrt{n} \cdot \log \frac{mn}{\delta}\right)$ that grows logarithmically in m. This matches the result of Alweiss et al. [2] in the noiseless setting, and is obtained by a comparatively simpler induction-based analysis. To establish this result, we first bound discrepancy in the online vector balancing problem (Theorem 6), and then bound multicolored discrepancy in an online setting with noisy vectors (Theorem 7). A known reduction from envy minimization to multicolored discrepancy minimization then yields the final result (Theorem 5).

1.2 Related Work

While most fair division literature assumes access to exact valuation functions, recognizing the practical difficulty of eliciting precise numbers has spurred interest in ordinal fair division. A central concept here is distortion [34], which quantifies the worst-case loss of a cardinal objective resulting from limited ordinal information. Distortion has been extensively studied across fair division [3, 27], matching [1, 5, 6, 24], voting [7, 18], and participatory budgeting [14] (see Anshelevich et al. [8] for a survey).

Closer in spirit to our setting are works that elicit ordinal feedback (often via pairwise comparisons) while allowing the underlying cardinal valuations to be chosen adversarially, e.g., [3, 4, 16, 25, 27]. Our results for stochastic true valuations also relate to the rich literature on stochastic fair division [11, 15, 22]. Finally, Bai et al. [12] study a *smoothed* model, where a worst-case utility profile is randomly perturbed to form the true valuations. In contrast, our model keeps true valuations fixed, while the perturbed profile serves as the observed, noisy input.

[3] An allocation $\mathcal{A} = (A_1, A_2, \ldots, A_n)$ is called balanced if all agents get roughly equal number of items, i.e., $|A_i| - |A_j| \leq 1$ for all $i, j \in [n]$.

[4] The revelation of information (preference) that governs the objective (envy), following a decision of the algorithm (allocation of an item) is also a key theme in areas like online convex optimization, the experts problem, and others.

[5] Alweiss et al. [2] study the problem of online discrepancy minimization in the absence of any noise.

2 Preliminaries

We consider the problem of dividing a set of m indivisible items among a set of $n \geq 2$ agents. We will represent the set of agents by $[n]$ and the set of items by $[m]$. Each agent $i \in [n]$ has a valuation function $v_i^* : 2^{[m]} \to \mathbb{R}$, which maps every subset of items $S \subseteq [m]$ to $v_i^*(S)$, the agent's value for S. In this paper, we focus on the case of *additive* valuation functions. That is, each agent $i \in [n]$ has a value $v_{i,j}^* \in \mathbb{R}$ for each item $j \in [m]$, and the value of agent i for a subset of items $S \subseteq [m]$ is $v_i^*(S) = \sum_{j \in S} v_{i,j}^*$.

Allocations. An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is an n-partition of $[m]$, where A_i denotes the set of items allocated to agent $i \in [n]$. That is, in any allocation $\mathcal{A}$, all items are allocated, $\bigcup_{i \in [n]} A_i = [m]$, and each item is allocated to only one agent, $A_i \cap A_j = \emptyset$ for all distinct agents $i, j \in [n]$.

Fairness Notions. We are interested in finding allocations that are fair, i.e., each agent is satisfied with the items they receive. Envy-freeness is the central fairness notion in fair division.

Definition 1 (Envy-freeness)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is* envy-free (EF) *if each agent prefers their own bundle over the bundle of any other agent, i.e., for all agents $i, j \in [n]$, we have $v_i^*(A_i) \geq v_i^*(A_j)$.*

Notably, envy-free allocations do not always exist when the items being allocated are indivisible. Hence, our goal in such scenarios is to minimize the maximum envy between any pair of agents. For any two agents $i, j \in [n]$, the envy that agent i has for agent j is defined as $v_i^*(A_j) - v_i^*(A_i)$. Our objective will be to minimize the maximum envy, i.e., $\max_{i \neq j} v_i^*(A_j) - v_i^*(A_i)$. In an envy-free allocation, the maximum envy is non-positive.

Inaccurate Preference Model. Our paper focuses on the setting where we do not get to observe the true values $v_{i,j}^*$ of the agents. Instead, we assume that for each agent $i \in [n]$ and item $j \in [m]$, we only have access to a noisy estimate $\hat{v}_{i,j}$ of the true values $v_{i,j}^*$. Indeed, the envy between the agents is still based on the true values $v_{i,j}^*$.

3 Stochastic True Valuations

In this section, we focus on the setting where the true valuations of all the agents, $\{v_{i,j}^*\}_{i,j}$, are stochastic. Section 3.1 describes our result for the case when the noise $(\hat{v}_{i,j} - v_{i,j}^*)$ is stochastic and Section 3.2 describes the result for worst-case, bounded noise.

3.1 Stochastic Values and Stochastic Noise

Let $\mathcal{D}$ and $\mathcal{D}'$ be two real-valued distributions. We consider a model where for every agent $i \in [n]$, item $j \in [m]$, we have $v_{i,j}^* \sim \mathcal{D}$, $\eta_{i,j} \sim \mathcal{D}'$, and the algorithm gets to observe values $\hat{v}_{i,j} = v_{i,j}^* + \eta_{i,j}$. We further assume that the distributions $\mathcal{D}$ and $\mathcal{D}'$ are unknown to the algorithm. In this setting, we prove that maximizing welfare with respect to the observed values—a simple and computationally efficient algorithm—outputs an envy-free allocation, with high probability.

Theorem 1. *Let $\mathcal{D}$ and $\mathcal{D}'$ be two fixed bounded distributions such that $\mathcal{D}$ is bounded, continuous, and has positive variance ($\mathcal{D}'$ can be arbitrary). Assume that $v_{i,j}^* \sim \mathcal{D}$ for all $i \in [n]$, $j \in [m]$, and we observe the noisy estimates $\hat{v}_{i,j} = v_{i,j}^* + \eta_{i,j}$ where $\eta_{i,j} \sim \mathcal{D}'$ for all $i \in [n]$, $j \in [m]$. Then, running the welfare maximization algorithm on the observed noisy values $\{\hat{v}_{i,j}\}_{i,j}$, computes an envy-free allocation with probability at least $1 - O(1/m)$ if $m/\log m \in \Omega(n)$.*

Theorem 1 only requires $\mathcal{D}$ to be a bounded continuous distribution with a positive variance; $\mathcal{D}'$ can be an arbitrary (discrete or continuous) distribution. It is easy to see that positive variance for $\mathcal{D}$ is a necessary requirement for envy-freeness. If $\mathcal{D}$ has zero variance, all item values will the identical with probability 1, so if m is not a multiple of n, an envy-free allocation does not exist. Continuity of $\mathcal{D}$ is assumed for the ease of exposition; however, it is not strictly necessary as long as ties are broken randomly.

To prove this result, we require a strict correlation inequality:

Lemma 1 (Theorem 1.2 of [29]). *Given any two real-valued, nondecreasing functions f, g and a real-valued random variable X, we have $\mathbb{E}[f(X) \cdot g(X)] > \mathbb{E}[f(X)] \cdot \mathbb{E}[g(X)]$ if and only if $\mathrm{Var}(f(X)) > 0$ and $\mathrm{Var}(g(X)) > 0$.*

Lemma 1 is a refinement of the well-known Chebyshev's association inequality:

Lemma 2 (Chebyshev's Association Inequality [17]). *Given any two real-valued, nondecreasing functions f, g and a real-valued random variable X, $\mathbb{E}[f(X) \cdot g(X)] \geq \mathbb{E}[f(X)] \cdot \mathbb{E}[g(X)]$.*

We note that while Lemma 2 is a classical result in probability theory, the general conditions for equality in Lemma 1 are not nearly as well known. We refer readers to the recent work of Jakubowski [29] which derives Lemma 1 and provides a relevant historical recount.

The strict inequality in Lemma 1 is used to prove Lemma 3, which plays a crucial role in establishing the main theorem of this section, Theorem 1. At a high-level, Lemma 3 shows that performing welfare maximization on inaccurate values is, in fact, effective in maximizing welfare with respect to the true valuations.

Lemma 3. *Let $\mathcal{D}$ and $\mathcal{D}'$ be two probability distributions. Furthermore, suppose $X_1, \ldots, X_n$, $Y_1, \ldots, Y_n$ are independent random variables such that each $X_i \sim \mathcal{D}$ and each $Y_i \sim \mathcal{D}'$. Then, (1) if $\mathcal{D}'$ has positive variance, $\mathbb{E}[X_1 \mid X_1 + Y_1 > \max_{i=2}^n (X_i + Y_i)] \geq \mathbb{E}[X_1 \mid X_1 + Y_1 \leq \max_{i=2}^n (X_i + Y_i)]$; and (2) if $\mathcal{D}$ has positive variance, $\mathbb{E}[X_1 \mid X_1 + Y_1 > \max_{i=2}^n (X_i + Y_i)] > \mathbb{E}[X_1 \mid X_1 + Y_1 \leq \max_{i=2}^n (X_i + Y_i)]$.*

The proofs of Theorem 1 and Lemma 3 are deferred to Appendix B.

3.2 Stochastic Values and Worst-Case Noise

We now consider a general setting where the value vector for each item j, $v_j^* = (v_{1,j}^*, \ldots, v_{n,j}^*)$ is drawn i.i.d. from a bounded vector-valued distribution $\mathcal{D}$ over $[0, b]^n$ for some bound b. This framework allows the agents' valuations for a specific item to be arbitrarily correlated (as the components of a single draw from $\mathcal{D}$). However, values across items are independent.

We show that if $\mathcal{D}$ satisfies a mild condition—essentially that no two agents are "identical" up to scaling—and if the number of items m is sufficiently large, an efficient algorithm can use noisy estimates to find an envy-free allocation with high probability. The condition is the following: Let $X = (X_1, \ldots, X_n) \sim \mathcal{D}$ be a random vector. We require that $\Pr[X_i = c \cdot X_j] < 1$ for all $i \neq j$ and constants $c \in \mathbb{R}$.

Theorem 2. *Let $\mathcal{D}$ be a bounded nonnegative distribution satisfying the non-identical condition. Assume that $v_j^* \sim \mathcal{D}$, independently, for all items j. For each $i \in [n]$, $j \in [m]$, let $\hat{v}_{i,j}$ be a noisy estimate of $v_{i,j}^*$ such that for some $\varepsilon > 0$, we have $\max_{i,j} |v_{i,j}^* - \hat{v}_{i,j}| \leq \varepsilon$. There is a constant $c_{\mathcal{D}}$ depending only on $\mathcal{D}$ and an algorithm such that if $\varepsilon \leq c_{\mathcal{D}}$, given the noisy estimates $\hat{v}_{i,j}$ for each i, j, the algorithm outputs an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that $\mathcal{A}$ is envy-free with probability at least $1 - \exp(-\Omega_{\mathcal{D}}(m))$.*

The algorithm uses linear programming to find a strongly envy-free fractional allocation with respect to the noisy estimates $\hat{v}$. It then applies randomized rounding to this fractional allocation to obtain the final integral allocation. The size of the linear program is polynomial in n and m, with coefficients based on $\hat{v}$. Thus, under the mild assumption that the estimates are polynomial in bit length (which must be the case if they were produced by a polynomial time algorithm), our algorithm runs in polynomial time.

The proof of Theorem 2 relies on establishing the existence of a strongly envy-free allocation in the underlying noiseless setting, which we formalize in the following lemma. This "strong" guarantee provides a buffer that is robust to the two sources of error: the bounded noise ε in the estimates and the error introduced by randomized rounding. This ensures that the final allocation is envy-free with high probability. This existence result may be of independent interest.

Lemma 4. *Let $\mathcal{D}$ be a distribution satisfying the above conditions. Assume that $v_j^* \sim \mathcal{D}$, independently, for all items j. Then, there is a constant $c'_{\mathcal{D}}$ depending only on $\mathcal{D}$ such that with probability at least $1 - \exp(-\Omega_{\mathcal{D}}(m))$, there exists an allocation $\mathcal{A}^*$ such that, for all agents $i, j \in [n]$, $v_i^*(A_i^*) \geq v_i^*(A_j^*) + c'_{\mathcal{D}}\, m$.*

The proofs of Lemma 4 and Theorem 2 are deferred to Appendix C.

4 Worst-Case True Valuations

The focus of this section will be on settings where the true preferences of the agents $v^*_{i,j}$ are fixed (i.e., worst-case). In Section 4.1, we provide optimal bounds for worst-case true values with worst-case noise. This is achieved by the simple Round-Robin algorithm. Section 4.2 presents two applications of this bound. Finally, in Section 4.3, we describe our results on worst-case values with stochastic noise using discrepancy and envy minimization.

4.1 Worst-Case Values and Worst-Case Noise

We prove that, given worst-case true valuations and (bounded) worst-case noise, we can find an allocation with maximum envy at most m/n times the noise (Theorem 3); and, we provide a matching lower bound (Theorem 4). The proofs are deferred to Appendix D.

The guarantee of Theorem 3 is achieved by the simple Round-Robin process. We generalize this by allowing each agent an arbitrary shift δ_i, meaning their estimated values approximate the shifted true values within an error of ε. This extension supports our later applications, where relying on shifted estimates is more realistic. The analysis remains unaffected because the Round-Robin algorithm depends only on the ordinal ranking of items, which is invariant under constant shifts.

Theorem 3 (Upper bound for worst-case noise)**.** *For every $i \in [n]$ and $j \in [m]$, let $\hat{v}_{i,j}$ be an estimate of $v^*_{i,j} \in [0, b]$, such that $\max_{i,j} |v^*_{i,j} - \hat{v}_{i,j} - \delta_i| \leq \varepsilon$, for some possibly unknown constants $\varepsilon > 0$ and $\delta_i \geq 0$. Given $\hat{v}_{i,j}$ for each i, j, we can efficiently compute an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that, for every pair of agents $i, i' \in [n]$, we have $v^*_i(A_{i'}) - v^*_i(A_i) \leq 2\varepsilon \left\lceil \frac{m}{n} \right\rceil + b$.*

Next, we prove that the envy bound achieved by Round-Robin is tight (up to an additive term) for deterministic algorithms.

Theorem 4 (Lower bound for worst-case noise)**.** *For every $i \in [n]$ and $j \in [m]$, let $\hat{v}_{i,j}$ be an estimate of $v^*_{i,j}$, such that $\max_{i,j} |v^*_{i,j} - \hat{v}_{i,j}| \leq \varepsilon$, for some $\varepsilon > 0$. No deterministic algorithm that takes $\hat{v}_{i,j}$s as input, and outputs a complete allocation, can guarantee that the envy between any two agents is strictly less than $2\varepsilon \frac{m}{n}$.*

4.2 Applications of Theorem 3

4.2.1 Estimating Valuations Via a BTL Query Model

We now describe a method for estimating agents' valuations under the assumption that each agent's preferences are generated according to an underlying Bradley-Terry-Luce (BTL) model. In this model, each agent is asked pairwise comparison queries of the form "which item do you like more, k or ℓ?" and they respond with a binary choice from $\{k, \ell\}$. We further assume the set of queried item pairs for each agent is determined independently by an Erdős–Rényi random graph where each unordered pair $\{k, \ell\}$ is included independently with probability p. For every queried pair, we collect K independent comparisons (i.e., K repeated queries per observed pair). A detailed formulation of this model is provided in Appendix A. This is a well-studied model for which statistical estimation of underlying BTL parameters is well understood [20]. Using the collected responses, we run a standard maximum-likelihood estimator for each agent $i \in [n]$, to recover $\tilde{\theta}_i \in \mathbb{R}^m$, an approximation of θ^*_i, which is a centered version of the true preference vector of agent i. Let $\Theta^* = [\theta^*_1; \ldots; \theta^*_n]^{\mathrm{T}}$ and $\hat{\Theta} = [\tilde{\theta}_1; \ldots; \tilde{\theta}_n]^{\mathrm{T}}$ denote the concatenated true and estimated preference matrices, then the following result provides estimation error bounds for each of θ^*_i.

Lemma 5 (ℓ^2 and ℓ^∞-Error Bounds [20, Theorem 3.1])**.** *Assume $p \geq c_0 \frac{\log m}{m}$ for a sufficiently large constant $c_0 > 0$, and that K comparisons are collected per observed pair under an Erdős–Rényi observation graph with parameter p. Let $\theta^* \in \mathbb{R}^m$ satisfies $\|\theta^*\|_\infty \leq b$ and ${\theta^*}^{\mathrm{T}} \mathbf{1}_m = 0$. Then, there exists a polynomial-time estimator that computes $\tilde{\theta} \in \mathbb{R}^m$ by asking $\binom{m}{2} pK$ average number of pairwise comparisons and satisfies the following bounds*

$$\|\tilde{\theta} - \theta^*\|_2 \leq c_2 \cdot \frac{1}{\sqrt{pK}}, \qquad \|\tilde{\theta} - \theta^*\|_\infty \leq c_\infty \cdot \sqrt{\frac{\log m}{mpK}},$$

with probability at least $1 - O(1/m^7)$, for some constants $c_2, c_\infty > 0$ only depending on b.

When the estimated values obtained from Lemma 5 are used as input to the Round-Robin algorithm, we obtain an allocation $\mathcal{A} = (A_1, A_2, \ldots, A_n)$ with a bounded envy. We formally state this in Application 1.

It is worth emphasizing that combining BTL estimation with Round-Robin succeeds specifically because Round-Robin produces *balanced* allocations. Under the BTL model, the pairwise comparison probabilities are invariant to additive shifts in an agent's valuations.[6] Consequently, the valuation can only be recovered up to an additive constant for each agent. However, since Round-Robin allocates roughly an equal number of items to each agent, these additive shifts cancel out, enabling us to obtain envy guarantees.

Application 1 (Inference and Fair Division)**.** *Assume there are m indivisible items and n additive agents with $n < m$, where $v^*_{i,j} \in [0,1]$ is the value of agent i for item j. Let $V^* \in [0,1]^{n\times m}$ be the valuation matrix where $V^*_{i,j} = v^*_{i,j}$. If agents respond to BTL-based pairwise queries, then there is an algorithm that asks $\binom{m}{2}pK$ average number of pairwise-comparison queries to each agent $i \in [n]$, and with probability at least $1 - O(1/m^6)$, produces an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ in polynomial time that has bounded maximum envy. Formally, the obtained allocation $\mathcal{A}$ satisfies $\max_{i,j} v_i(A_j) - v_i(A_i) = O\left(\sqrt{\frac{m \log m}{n^2 pK}}\right)$, where K is the number of queries per-pair and p is the observation graph probability satisfying $p = \Omega\left(\frac{\log m}{m}\right)$.*

The proof of Application 1 can be found in Appendix E.

4.2.2 Worst-Case Values and Stochastic Noise

We consider the following simple model for stochastic noise: $\hat{v}_{i,j} = v^*_{i,j} + \varepsilon_{i,j}$, for $\varepsilon_{i,j} = \sigma_{i,j} \cdot z_{i,j}$. We assume that $z_{i,j}$s are drawn i.i.d. from a non-negative distribution $\mathcal{D}$, and $\sigma_{i,j}$s are drawn uniformly at random from $\{-1, 1\}$. That is, $z_{i,j}$ is the magnitude of the noise, and $\sigma_{i,j}$ denotes whether the noise underestimates or overestimates the value. The next theorem shows that, when $\mathcal{D}$ is not heavy tailed, then we can efficiently compute an allocation with small envy (with respect to the true valuations). Formally, our condition on $\mathcal{D}$ is that it has monotone hazard rate (MHR): $\frac{1-F(x)}{f(x)}$ is a non-increasing function, where $F(x)$ and $f(x)$ are its cumulative distribution function and probability density function, respectively. We defer the proof of this application to Appendix E.

Application 2. *For every $i \in [n]$ and $j \in [m]$, let $z_{i,j}$s be i.i.d. draws from a non-negative MHR distribution $\mathcal{D}$ such that $\mathbb{E}[\mathcal{D}] \le \frac{n}{m \log(nm)}$, and $\sigma_{i,j}$s be uniformly random draws from $\{-1, 1\}$. Let $\hat{v}_{i,j} = v^*_{i,j} + \varepsilon_{i,j}$, for $\varepsilon_{i,j} = \sigma_{i,j} \cdot z_{i,j}$. Given $\hat{v}_{i,j}$ and assuming $v^*_{i,j} \in [0, b]$ for each i, j, we can efficiently compute an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that, for every pair of agents $i, i' \in [n]$, we have $v^*_i(A_{i'}) - v^*_i(A_i) \le 2 \log(nm)\, \mathbb{E}[\mathcal{D}] \left\lceil \frac{m}{n} \right\rceil + b$, with probability at least $1 - \frac{1}{(nm)^{3/5}}$.*

4.3 Discrepancy Minimization with Inaccurate Inputs

Here, we focus on a variant of the multicolored discrepancy minimization problem. Upper bounds on multicolored discrepancy are derived, as it is known that these bounds directly translate to the problem of envy minimization [2, 28]. We begin with a brief refresher on discrepancy minimization.

In the vector balancing problem, given a set of vectors $u_1, u_2, \ldots, u_m$ satisfying $\|u_i\|_\infty \le 1$, the goal is to compute signs $s_1, \ldots, s_m \in \{-1, 1\}$ to minimize the *discrepancy*, defined as $\|\sum_{i=1}^m s_i u_i\|_\infty$. The multicolored discrepancy minimization problem is a generalization of the vector balancing problem in which we have to assign one color, among a set of $[k] = \{1, 2, \ldots, k\}$ colors, to each vector, to minimize the *multicolored discrepancy*, i.e., $\max_{i,j\in[k]} \|\sum_{v\in S_i} v - \sum_{v\in S_j} v\|_\infty$, where for each $\ell \in [k]$, S_ℓ is the set of vectors that are assigned the color ℓ.

To see how envy minimization reduces to multicolored discrepancy minimization, consider the following. If we let each agent $i \in [n]$ correspond to a color (i.e., we set $k = n$) and each item $j \in [m]$ correspond to a vector $v^*_j = (v^*_{1,j}, v^*_{2,j}, \ldots, v^*_{n,j})/\sqrt{n}$ (ensuring $\|v^*_j\|_\infty \le 1$), then we get an instance of the multicolored discrepancy minimization problem. Moreover, for any allocation $\mathcal{A} = (A_1, \ldots, A_n)$, the maximum envy $\max_{i,j\in[n]} v_i(A_j) - v_i(A_i)$ is always at most

[6] Formally, the pairwise comparison probability in BTL, stated in Equation (1), depends only on the difference in preference values and not their absolute values.

the multicolored discrepancy $\max_{i,j\in[n]} \| \sum_{v\in S_i} v - \sum_{v\in S_j} v\|_\infty$ of the corresponding problem, where the set S_i contains the vectors that correspond to the bundle of items A_i assigned to agent i. Hence, $\max_{i,j\in[n]} v_i(A_j) - v_i(A_i) \leq \max_{i,j\in[n]} \| \sum_{v\in S_i} v - \sum_{v\in S_j} v\|_\infty$ and any upper bound on multicolored discrepancy translates to an upper bound on maximum envy.

We study envy minimization for a model in which we first observe the inaccurate preferences $(v_j^* + \eta_j) \in \mathbb{R}^n$ for each item $j \in [m]$. We then allocate this item to an agent, and after the allocation, the true preference vector v_j^* is revealed to us. We assume that $v_{i,j}^* \in [-1, 1]$ and $\eta_{i,j} \in [-\varepsilon, \varepsilon]$ are both worst-case.[7] In this model, we give an efficient *online* algorithm that computes an allocation whose maximum envy is upper bounded by $O\left(\sqrt{n}(\log \frac{mn}{\delta} + m\varepsilon)\right)$ with probability at least $1 - \delta$.

Theorem 5. *Let $v_1^* + \eta_1, v_2^* + \eta_2, \ldots, v_m^* + \eta_m \in \mathbb{R}^n$ be a sequence of items (valuation vectors) that arrive one at a time, where $v_{i,j}^* \in [-1, 1], \eta_{i,j} \in [-\varepsilon, \varepsilon]$ for all $i \in [n], j \in [m]$. When each item $v_i^* + \eta_i$ arrives, we need to allocate it to one of the n agents, after which the true valuation vector v_i^* is revealed to us. In this model, there is an efficient online algorithm that computes an allocation $\mathcal{A} = (A_1, A_2, \ldots, A_n)$ that guarantees the following maximum envy bound $\max_{i,j\in[n]} v_i^*(A_j) - v_i^*(A_i) \leq \frac{27\sqrt{n}}{C} \log \frac{5nm}{\delta} + 6m\sqrt{n}\varepsilon$, with probability at least $1 - \delta$ for a universal constant $C > 0$.*

To prove Theorem 5, we first develop an online algorithm for vector balancing problem with noisy input vectors (Section 4.3.1). Using this result, we obtain a bound for the corresponding multicolored discrepancy problem (Section 4.3.2). Finally, via the reduction from envy minimization to multicolored discrepancy minimization explained earlier, we directly obtain Theorem 5.

4.3.1 Vector Balancing with Noisy Input

We consider the following weighted vector-balancing problem with inaccurate vectors as input. Formally, given a sequence of inaccurate vectors $v_1^* + \eta_1, v_2^* + \eta_2, \ldots, v_m^* + \eta_m$ as input, we want to find signs $s_1, s_2, \ldots, s_m \in \{p, p-1\}$ where $p \in [1/2, 2/3]$ is a fixed parameter, to minimize $\| \sum_{i=1}^m s_i v_j^* \|_\infty$. We show that $\| \sum_{i=1}^m s_i v_j^* \|_\infty \leq O\left(\sqrt{n}(\log \frac{mn}{\delta} + m\varepsilon)\right)$ with probability at least $1 - \delta$ is achieved by a modified version of the online algorithm given by [2], which achieves a similar logarithmic bound on discrepancy, in the absence of noise. Our algorithm (Balance Under Noise) is stated as Algorithm 1.

Algorithm 1: BALANCE UNDER NOISE

Input : Parameter $p \in [1/2, 2/3]$ and a sequence of vector $v_1^* + \eta_1, v_2^* + \eta_2, \ldots, v_m^* + \eta_m$ where $\|v_t^*\|_\infty \leq 1, \|\eta_t\|_\infty \leq \varepsilon$ for all $t \in [m]$.

Output : Signs $s_1, s_2, \ldots, s_m \in \{p, p-1\}$.

Initialize $w_1 \leftarrow \mathbf{0} \in \mathbb{R}^n$ and $s_1, s_2, \ldots, s_m \leftarrow 0$.
Define function $r(x) := \min\{1, \max\{0, x\}\}$ for all $x \in \mathbb{R}$.

for $t \leftarrow 1, 2, \ldots, m$ **do**

Set $c_t \leftarrow \frac{9n}{4C} \log \frac{5nm}{\delta} + \varepsilon nt$

Randomly sample $s_t \sim \begin{cases} p, & \text{with probability } r\left((1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}\right), \\ p-1, & \text{with probability } 1 - r\left((1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}\right). \end{cases}$

Set $w_{t+1} \leftarrow w_t + s_t v_t^*$

end

return *signs* $s_1, s_2, \ldots, s_m$.

Algorithm 1 maintains a running discrepancy vector, $w_t = \sum_{j=1}^{t-1} s_j v_j^*$. When a new inaccurate vector $v_t^* + \eta_t$ arrives, it simply samples the sign s_t from the following distribution: p with probability $r((1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t})$ and $p-1$ otherwise (Line 1). The function $r(\cdot)$ defined in Line 1, ensures that the probability value is valid, i.e., lies between zero and one. Note that the probability is computed

[7] This is without loss of generality, as the bounds we obtain can be scaled by a factor of b if $v_{i,j}^* \in [-b, b]$ and $\eta_{i,j} \in [-\varepsilon b, \varepsilon b]$.

based on the inaccurate vector $v_t^* + \eta_t$. Once the sign s_t has been sampled, the true vector v_t^* is revealed, and w_{t+1} is updated (Line 1).

We obtain the following guarantee about vector balancing, its proof appearing in Appendix F.1.

Theorem 6. *During the execution of Algorithm 1, the random vector $w_t = \sum_{i=1}^{t-1} s_i v_i^*$ satisfies $\|w_t\|_\infty \leq \frac{9\sqrt{n}}{4C} \log \frac{5nm}{\delta} + \varepsilon t \sqrt{n}$, for all $t \in [m]$ with probability at least $1 - \delta$ for a universal constant $C > 0$.*

4.3.2 Multicolored Discrepancy and Envy Minimization

We now generalize Theorem 6 to the multicolored discrepancy problem. The idea is based on a known reduction utilized in prior work [2, 28].

We begin by first describing the algorithm that achieves the multicolored discrepancy bound of Theorem 7. Consider a complete and full binary tree $\mathcal{T}$ having exactly k leaves and root node r.[8] For each internal node u of $\mathcal{T}$, we use n_ℓ^u and n_r^u to denote the number of leaves in its left and right subtree, respectively. We run a copy of Algorithm 1 at each internal node u of $\mathcal{T}$ by setting the parameter p to $p_u = n_\ell^u / (n_\ell^u + n_r^u)$.

When each vector $v_j^* + \eta_j$ arrives, we pass it down along the tree from the root to one of the leaves; $v_i^* + \eta_j$ is then assigned a color equal to the index of the leaf it eventually arrives at. In particular, first $v_j^* + \eta_j$ is passed through the copy of Algorithm 1 running at the root r of $\mathcal{T}$. If $v_j^* + \eta_j$ is assigned a sign of p_r by Algorithm 1 running at r, then it is passed down to the next node in the right subtree, if $v_j^* + \eta_j$ is assigned a sign of $p_r - 1$, then it is passed down to the next node in the left subtree. Suppose the next node is u, then we again check whether $v_j^* + \eta_j$ is assigned a sign of p_u, $p_u - 1$ at u, and pass it down to the right, left subtree, respectively. This continues until the vector $v_j^* + \eta_j$ reaches a leaf node, i.e, it gets assigned a color equal to the index of the leaf node. When this occurs, the true vector v_j^* is revealed. Using v_j^*, the value of the vector w_j^u (superscript u identifies the node) used in Algorithm 1 is updated at each internal node through which $v_j^* + \eta_j$ passes; this update occurs in Line 1. This process of assigning vectors to leaves is repeated for all the vectors. This completes the description of the multicolored discrepancy algorithm.

The guarantee of this algorithm is stated in Theorem 7, its proof appears in Appendix F.2. Notably, while we run Algorithm 1 in every internal node of $\mathcal{T}$ having a depth of tree $O(\log k)$, the final multicolored discrepancy bound we obtain is only a constant factor larger than Theorem 6.

Theorem 7. *Let $k \geq 2$ be an integer and $v_1^* + \eta_1, v_2^* + \eta_2, \ldots, v_m^* + \eta_m \in \mathbb{R}^n$ be a sequence of vectors that arrive sequentially, where $\|v_i^*\|_\infty \leq 1, \|\eta_i\|_\infty \leq \varepsilon \leq 1$ for all $i \in [m]$. When each vector $v_i^* + \eta_i$ arrives, we need to assign it a color from $[k]$ after which the vector v_i^* is revealed to us. In this model, there is an efficient algorithm that achieves a multicolored discrepancy bound of $\max_{i,j \in [k]} \| \sum_{v \in S_i} v - \sum_{v \in S_j} v \|_\infty \leq \frac{27\sqrt{n}}{2C} \log \frac{5nmk}{\delta} + 6\varepsilon m \sqrt{n}$, with probability at least $1 - \delta$, where S_a is the set of vectors that are assigned the color $a \in [k]$ after all the vectors have arrived.*

Finally, we obtain Theorem 5 by setting $k = n$ in Theorem 7 and recalling that any upper bound on multicolored discrepancy is also an upper bound for the corresponding envy-minimization problem, as highlighted at the start of Section 4.3.

5 Conclusion

This paper establishes a practical framework for allocating indivisible items using noisy cardinal preferences. We show that algorithms relying on imprecise estimates achieve fairness guarantees comparable to exact-value models, removing the requirement for perfect numerical precision. Specifically, standard welfare maximization yields envy-free allocations with high probability for stochastic valuations. Under worst-case bounded noise, the Round-Robin algorithm achieves the optimal maximum envy bound, a result directly applicable to BTL preference estimation. For online settings where true values emerge only post-allocation, we introduce an algorithm that maintains a logarithmic envy bound by generalizing discrepancy theory for noisy inputs. Ultimately, these results prove that

[8] A binary tree is *full* if every internal node has either 0 or 2 children. A binary tree is *complete* if all levels except possibly the last one are completely filled.

additive fair division frameworks are highly robust to preference inaccuracies, enabling realistic, low-burden allocation systems.

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A Preliminaries for the BTL Model

This appendix provides the formal setup for agents' preference elicitation via the BTL model. The material here expounds on the estimation procedure described in Section 4.2.

We consider a natural extension of the well-studied BTL model [19, 31, 33], where each agent has a separate set of BTL preferences. Specifically, for n agents and m items, the model assumes that each agent $i \in [n]$ has an associated (unknown) preference score $\theta^*_{i,j} \in [-b, b]$ for every item $j \in [m]$. Under the BTL model, the probability of agent i preferring item j over item k depends only on the difference in their latent scores and is given by:

$$\mathbb{P}(\text{agent } i \text{ prefers item } j \text{ over } k) = \frac{\exp(\theta^*_{i,j})}{\exp(\theta^*_{i,j}) + \exp(\theta^*_{i,k})} = \psi(\theta^*_{i,j} - \theta^*_{i,k}), \tag{1}$$

where $\psi(t) = 1/(1 + \exp(-t))$ denotes the sigmoid function.

We collect the preference scores as a matrix $\Theta^* \in \mathbb{R}^{n \times m}$ such that $\Theta^*_{i,j} = \theta^*_{i,j}$. Our goal is to estimate the unknown preference matrix Θ^* from noisy pairwise comparison queries collected from the agents. Since the preference probabilities in Eq. (1) are invariant to translation, we impose the following identifiability constraint

$$\Theta^* \mathbf{1}_m = \mathbf{0}_n. \tag{2}$$

For each agent, comparisons are observed only on a subset of item pairs. This comparison structure is modeled by an agent-specific undirected graph $\mathcal{E}_i$. Specifically, for each agent $i \in [n]$, an edge $(j, k) \in \mathcal{E}_i$ if and only if agent i provides a comparison between items j and k. We assume that the observation graph of each agent is drawn independently from an Erdős-Rényi random graph model with parameter p.

We assume that for every agent $i \in [n]$ and for each pair $(j, k) \in \mathcal{E}_i$ with $k > j$, we observe K independent comparisons. Let $y^{(i,l)}_{j,k} \in \{0, 1\}$ denote the outcome of the l-th comparison between items j and k by agent i, where $y^{(i,l)}_{j,k} = 1$ indicates a preference for item j over item k. Under the BTL model, these outcomes are generated according to

$$y^{(i,l)}_{j,k} = \begin{cases} 1, & \text{with probability } \psi(\theta^*_{i,j} - \theta^*_{i,k}) \\ 0, & \text{otherwise} \end{cases}, \tag{3}$$

For convenience, we set $y^{(i,l)}_{k,j} = 1 - y^{(i,l)}_{j,k}$ for all $(k > j)$ with $(j, k) \in \mathcal{E}_i$. We define the empirical preference probabilities $y^{(i)}_{j,k} = \frac{1}{K} \sum_{l=1}^{K} y^{(i,l)}_{j,k}$. Given the observed empirical preference probabilities $\{y^{(i)}_{j,k}\}_{i,j,k}$, our goal is to estimate the latent preference matrix Θ^*.

For each agent $i \in [n]$, we estimate the agent's parameters $\tilde{\theta}^{(i)}$ by solving the following marginal log-likelihood problem using only agent i's preference data (cf. [20]). This problem takes the form

$$\tilde{\theta}^{(i)} = \underset{\theta : \theta^{\mathrm{T}} \mathbf{1}_m = 0}{\arg\min} \left\{ -\frac{1}{mp} \sum_{(j,k) \in \mathcal{E}_i} y^{(i)}_{j,k} \log \psi(\theta_j - \theta_k) + (1 - y^{(i)}_{j,k}) \log(1 - \psi(\theta_j - \theta_k)) \right\}. \tag{4}$$

This convex optimization yields n vectors $\tilde{\theta}^{(i)} \in \mathbb{R}^m$, one for each agent. Stacking these vectors as rows produces the initial matrix estimate $\hat{\Theta} = \left[\tilde{\theta}^{(1)\mathrm{T}}; \tilde{\theta}^{(2)\mathrm{T}}; \ldots; \tilde{\theta}^{(n)\mathrm{T}}\right] \in \mathbb{R}^{n \times m}$. The estimation error bounds for this estimator are quantified in Lemma 5.

B Missing Proofs from Section 3.1

Lemma 3. *Let $\mathcal{D}$ and $\mathcal{D}'$ be two probability distributions. Furthermore, suppose $X_1, \ldots, X_n$, $Y_1, \ldots, Y_n$ are independent random variables such that each $X_i \sim \mathcal{D}$ and each $Y_i \sim \mathcal{D}'$. Then, (1) if $\mathcal{D}'$ has positive variance, $\mathbb{E}[X_1 \mid X_1 + Y_1 > \max_{i=2}^n (X_i + Y_i)] \geq \mathbb{E}[X_1 \mid X_1 + Y_1 \leq \max_{i=2}^n (X_i + Y_i)]$; and (2) if $\mathcal{D}$ has positive variance, $\mathbb{E}[X_1 \mid X_1 + Y_1 > \max_{i=2}^n (X_i + Y_i)] > \mathbb{E}[X_1 \mid X_1 + Y_1 \leq \max_{i=2}^n (X_i + Y_i)]$.*

Proof. Define $\mathcal{E}_>$ to be the event $[X_1 + Y_1 > \max_{i=2}^n X_i + Y_i]$, and $\mathcal{E}_\leq$ to be the event $[X_1 + Y_1 \leq \max_{i=2}^n X_i + Y_i]$.

(1) If $\mathcal{D}'$ has positive variance, then the random variables $X_i + Y_i$ have positive variance. Positive variance implies that $\mathbb{P}(\mathcal{E}_>), \mathbb{P}(\mathcal{E}_\leq) > 0$, i.e., the conditional expectations $\mathbb{E}[X_1 \mid \mathcal{E}_>], \mathbb{E}[X_1 \mid \mathcal{E}_\leq]$ are well defined. To prove (1), it suffices to show that

$$\mathbb{E}[X_1 \mid \mathcal{E}_>] \geq \mathbb{E}[X_1] = \mathbb{E}[X_1 \mid \mathcal{E}_>]\mathbb{P}(\mathcal{E}_>) + \mathbb{E}[X_1 \mid \mathcal{E}_\leq]\mathbb{P}(\mathcal{E}_\leq),$$

or equivalently,

$$\mathbb{E}[X_1 \mid \mathcal{E}_>] = \frac{\mathbb{E}[X_1 \mathbb{1}_{\mathcal{E}_>}]}{\mathbb{P}(\mathcal{E}_>)} \geq \mathbb{E}[X_1].$$

This can be written as

$$\mathbb{E}[X_1 \mathbb{1}_{\mathcal{E}_>}] \geq \mathbb{E}[X_1]\mathbb{E}[\mathbb{1}_{\mathcal{E}_>}].$$

Define $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x$ and $g : \mathbb{R}^2 \to \{0, 1\}$, $g(x, z) = \mathbb{1}\{x > z\}$. Additionally, let random variable $Z \coloneqq \max_{i=2}^n (X_i + (Y_i - Y_1))$; note that $\mathbb{1}_{\mathcal{E}_>} = \mathbb{1}[X_1 > Z]$. Then, we want to show that

$$\mathbb{E}[h(X_1)g(X_1, Z)] \geq \mathbb{E}[h(X_1)]\mathbb{E}[g(X_1, Z)].$$

To prove this, we condition on Z. Then, using the association inequality (Lemma 2) we have that

$$\mathbb{E}[h(X_1)g(X_1, Z) \mid Z] \geq \mathbb{E}[h(X_1) \mid Z]\mathbb{E}[g(X_1, Z) \mid Z] = \mathbb{E}[h(X_1)]\mathbb{E}[g(X_1, Z) \mid Z],$$

where we remove the conditioning in $\mathbb{E}[h(X_1) \mid Z]$ due to independence. Then, taking expectations with respect to Z gives us,

$$\mathbb{E}[h(X_1)g(X_1, Z)] \geq \mathbb{E}[h(X_1)]\mathbb{E}[g(X_1, Z)].$$

(2) Based on the previous part, in order to prove (2), we need to establish $\mathbb{E}[h(X_1)g(X_1, Z)] > \mathbb{E}[X_1]\mathbb{E}[g(X_1, Z)]$. Similar to the above, we condition on a fixed value on Z. Lemma 1 implies that conditioned on Z, the strict inequality

$$\mathbb{E}[h(X_1)g(X_1, Z) \mid Z] > \mathbb{E}[h(X_1)]\mathbb{E}[g(X_1, Z) \mid Z], \tag{5}$$

holds if and only if $\mathrm{Var}(h(X_1)) = \mathrm{Var}(X_1) > 0$ and $\mathrm{Var}(\mathbb{1}\{X_1 > Z\}) = \mathbb{P}[X_1 > Z](1 - \mathbb{P}[X_1 > Z]) > 0$; here, the former condition is true as per our assumption on $\mathcal{D}$, and the latter condition is equivalent to $\mathbb{P}[X_1 > Z] \in (0, 1)$. Hence, for a fixed value of Z, if $\mathbb{P}[X_1 > Z] \in (0, 1)$, then, and only then, we have $\mathbb{E}[h(X_1)g(X_1, Z) \mid Z] > \mathbb{E}[h(X_1)]\mathbb{E}[g(X_1, Z) \mid Z]$. The final step of the proof will be to show that the strict inequality persists after taking expectations with respect to Z. To argue this, we will show that the probability

$$\mathbb{P}_Z[\mathbb{P}_{X_1}[X_1 > Z] \in (0, 1)] > 0,$$

i.e., with positive probability over selecting Z, we get a strict inequality conditioned on Z, thereby ensuring that the final inequality $\mathbb{E}[h(X_1)g(X_1, Z)] > \mathbb{E}[X_1]\mathbb{E}[g(X_1, Z)]$ obtained after taking expectations with respect to Z is strict as well.

Let $\mathcal{E}$ be the event that $Y_1 \geq \max_{i=2}^n Y_2$. Since Y_is are iid drawn from $\mathcal{D}'$, we have $\mathbb{P}[\mathcal{E}] = \mathbb{P}[Y_1 \geq \max_{i=2}^n Y_i] \geq \frac{1}{n} > 0$. Additionally, conditioned on $\mathcal{E}$, we have $Z \leq \max_{i=2}^n X_i$. Hence,

$$\begin{aligned}
\mathbb{P}_Z[\mathbb{P}_{X_1}[X_1 > Z] \in (0, 1)] &\geq \mathbb{P}[\mathcal{E}] \cdot \mathbb{P}_Z[\mathbb{P}_{X_1}[X_1 > Z] \in (0, 1) \mid \mathcal{E}] \\
&\geq \mathbb{P}[\mathcal{E}] \cdot \mathbb{P}_Z[\mathbb{P}_{X_1}[X_1 > \max_{i=2}^n X_i] \in (0, 1) \mid \mathcal{E}] \\
&= \mathbb{P}[\mathcal{E}] \cdot \mathbb{P}_{X_2,\ldots,X_n}[\mathbb{P}_{X_1}[X_1 > \max_{i=2}^n X_i] \in (0, 1)],
\end{aligned}$$

where we drop the conditioning in the second term because X_is and Y_is are independent and $\mathcal{E}$ only depends on Y_is. Finally, note $\mathbb{P}[\mathcal{E}] > 0$, and the second term in the product above is also positive since the variance of $\mathcal{D}$ is positive. This implies that $\mathbb{P}_Z[\mathbb{P}_{X_1}[X_1 > Z] \in (0, 1)]$ is positive and completes the proof. □

Theorem 1. *Let $\mathcal{D}$ and $\mathcal{D}'$ be two fixed bounded distributions such that $\mathcal{D}$ is bounded, continuous, and has positive variance ($\mathcal{D}'$ can be arbitrary). Assume that $v^*_{i,j} \sim \mathcal{D}$ for all $i \in [n]$, $j \in [m]$, and we observe the noisy estimates $\hat{v}_{i,j} = v^*_{i,j} + \eta_{i,j}$ where $\eta_{i,j} \sim \mathcal{D}'$ for all $i \in [n]$, $j \in [m]$. Then, running the welfare maximization algorithm on the observed noisy values $\{\hat{v}_{i,j}\}_{i,j}$, computes an envy-free allocation with probability at least $1 - O(1/m)$ if $m/\log m \in \Omega(n)$.*

Proof. To allocate the items to the agents, we run the social welfare maximization algorithm on the noisy estimates, i.e., we allocate each item $j \in [m]$ to the agent $i \in \arg\max_{i\in[n]}\{\hat{v}_{i,j}\}$ (tie-breaking randomly). We assume that $\mathcal{D}$ is bounded within $[0, 1]$, and $\mathcal{D}'$ is scaled accordingly. This is without loss of generality as our algorithm is scalar invariant.

We observe that the algorithm satisfies the following two properties: first, for each item $j \in [m]$, we have $\mathbb{P}(i \in \arg\max_{i\in[n]}\{\hat{v}_{i,j}\}) = 1/n$, i.e., every agent has an equal, $1/n$, chance of getting each item. This follows from the fact that $\hat{v}_{i,j}$ are continuous random variables, and are independent and identically distributed.

Second, we have that $\mathbb{E}[v^*_{i,j} \mid i \in \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}] > \mathbb{E}[v^*_{i,j} \mid i \notin \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}]$, i.e., agents have a higher *true* value for an item they receive based on the *observed* noisy values. This is the most crucial property of our analysis, and it follows from Lemma 3 since $\mathcal{D}$ is continuous and has positive variance. With these two properties, we will show that this algorithm results in envy-free allocations with high probability if $m/\log m = \Omega(n)$.[9]

Define indicator random variables $X_{i,j} \coloneqq \mathbb{1}[i \in \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}]$, which are true if and only if agent i received item j. Define $\mathcal{B}_{i,\ell} = \mathbb{1}[\sum_{j=1}^m v_{i,j}X_{i,j} < \sum_{j=1}^m v_{i,j}X_{\ell,j}]$ to be the event that agent i envies agent ℓ after the execution of the algorithm. We want to argue that the probability $\mathbb{P}(\cup_{i\neq\ell}\mathcal{B}_{i,\ell})$ is tiny. Towards this, we begin by analyzing the $\mathbb{P}[\mathcal{B}_{i,\ell}]$.

For all $j \in [m]$, define $\mu_j \coloneqq \mathbb{E}[v_{i,j} \mid i \in \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}]$ and $\mu'_j \coloneqq \mathbb{E}[v_{i,j} \mid i \notin \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}]$. Note that as items behave identically, we have $\mu_1 = \mu_2 = \ldots = \mu_m$ and $\mu'_1 = \mu'_2 = \ldots = \mu'_m$, so we define $\mu = \mu_1$ and $\mu' = \mu'_1$. Furthermore, for fixed $\mathcal{D}$ and $\mathcal{D}'$, we know that $\mu > \mu'$ from part (2) of Lemma 3, since $\mathcal{D}$ and $\mathcal{D}'$ satisfy the required conditions.

Note that for the event $\mathcal{B}_{i,\ell}$ to occur, either the event $\mathcal{B}^1_{i,\ell} \coloneqq \mathbb{1}[\sum_{j=1}^m v_{i,j}X_{i,j} < \frac{m}{n}\left(\frac{\mu+\mu'}{2}\right)]$ or $\mathcal{B}^2_{i,\ell} \coloneqq \mathbb{1}[\sum_{j=1}^m v_{i,j}X_{\ell,j} > \frac{m}{n}\left(\frac{\mu+\mu'}{2}\right)]$ must occur; indeed, if $\mathcal{B}^1_{i,\ell}$ and $\mathcal{B}^2_{i,\ell}$ do not occur, then $\mathcal{B}_{i,\ell}$ cannot occur. Hence, to show that $\mathcal{B}_{i,\ell}$ occurs with small probability, we will show that $\mathcal{B}^1_{i,\ell}$ and $\mathcal{B}^2_{i,\ell}$ occur with small probability.

Note that $\mathbb{E}[\sum_{j=1}^m v_{i,j}X_{i,j}] = m \cdot \mathbb{P}(X_{i,j} = 1) \cdot \mathbb{E}[v_{i,j} \mid i \in \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}] = \frac{m\mu}{n}$ and $\mathbb{E}[\sum_{j=1}^m v_{i,j}X_{\ell,j}] = m \cdot \mathbb{P}(X_{\ell,j} = 1) \cdot \mathbb{E}[v_{i,j} \mid i \in \arg\max_{k\in[n]}\{\hat{v}_{k,j}\}] = \frac{m\mu'}{n}$. Using this, we can rewrite

$$\mathbb{P}(\mathcal{B}^1_{i,\ell}) = \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{i,j} < \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{i,j}]\left(1 - \frac{\mu-\mu'}{2\mu}\right)\right)$$

and

$$\mathbb{P}(\mathcal{B}^2_{i,\ell}) = \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{\ell,j} > \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{\ell,j}]\left(1 + \frac{\mu-\mu'}{2\mu'}\right)\right).$$

Define $\delta = \min\{\frac{\mu-\mu'}{2\mu}, 1\}$, and note this is a fixed quantity, as μ and μ' are fixed. Since $\sum_{j=1}^m v_{i,j}X_{i,j}$ is a nonnegative random variable, we get that

$$\begin{aligned}\mathbb{P}(\mathcal{B}^1_{i,\ell}) &= \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{i,j} < \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{i,j}]\left(1 - \frac{\mu-\mu'}{2\mu}\right)\right)\\ &= \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{i,j} < \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{i,j}]\,(1-\delta)\right).\end{aligned}$$

Additionally, the fact that $\mu > \mu'$ implies that

[9]The subsequent argument will follow a similar analysis presented in [22].

$$\mathbb{P}(\mathcal{B}^2_{i,\ell}) = \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{\ell,j} > \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{\ell,j}]\left(1+\frac{\mu-\mu'}{2\mu'}\right)\right)$$
$$\leq \mathbb{P}\left(\sum_{j=1}^m v_{i,j}X_{\ell,j} > \mathbb{E}[\sum_{j=1}^m v_{i,j}X_{\ell,j}]\,(1+\delta)\right).$$

Now, we can use the Chernoff bounds on the sum of independent-and-$[0,1]$-bounded variables $v_{i,j}X_{i,j}$ and $v_{i,j}X_{\ell,j}$ to obtain the following upper bound on $\mathbb{P}(\mathcal{B}_{i,\ell})$,

$$\begin{aligned}
\mathbb{P}(\mathcal{B}_{i,\ell}) &\leq \mathbb{P}(\mathcal{B}^1_{i,\ell} \cup \mathcal{B}^2_{i,\ell}) \\
&\leq \mathbb{P}(\mathcal{B}^1_{i,\ell}) + \mathbb{P}(\mathcal{B}^2_{i,\ell}) && \text{(union bound)} \\
&\leq \exp\left(-\mathbb{E}\left[\sum_{j=1}^m v_{i,j}X_{i,j}\right]\frac{\delta^2}{2}\right) + \exp\left(-\mathbb{E}\left[\sum_{j=1}^m v_{i,j}X_{\ell,j}\right]\frac{\delta^2}{3}\right) && \text{(Chernoff bound)} \\
&= \exp\left(-\frac{m\mu}{n}\cdot\frac{\delta^2}{2}\right) + \exp\left(-\frac{m\mu'}{n}\cdot\frac{\delta^2}{3}\right) < 2\exp\left(-\frac{m\mu'}{n}\cdot\frac{\delta^2}{3}\right).
\end{aligned}$$

For m where $m \geq \frac{9n\log m}{\mu'\delta^2}$, we get $\mathbb{P}(\mathcal{B}_{i,\ell}) \leq 2\exp(-3\log m) = 2/m^3$. Hence, via a union bound, we get that the probability of the resulting allocation not being envy-free is $\mathbb{P}(\cup_{i\neq\ell}\mathcal{B}_{i,\ell}) \leq \frac{2n^2}{m^3} \in O(1/m)$. This completes the proof. □

C Missing Proofs from Section 3.2

We now prove the full theorem, again restated here for convenience.

Theorem 2. *Let $\mathcal{D}$ be a bounded nonnegative distribution satisfying the non-identical condition. Assume that $v^*_j \sim \mathcal{D}$, independently, for all items j. For each $i \in [n]$, $j \in [m]$, let $\hat{v}_{i,j}$ be a noisy estimate of $v^*_{i,j}$ such that for some $\varepsilon > 0$, we have $\max_{i,j}|v^*_{i,j} - \hat{v}_{i,j}| \leq \varepsilon$. There is a constant $c_{\mathcal{D}}$ depending only on $\mathcal{D}$ and an algorithm such that if $\varepsilon \leq c_{\mathcal{D}}$, given the noisy estimates $\hat{v}_{i,j}$ for each i, j, the algorithm outputs an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that $\mathcal{A}$ is envy-free with probability at least $1 - \exp(-\Omega_{\mathcal{D}}(m))$.*

Proof of Theorem 2. Let $\mathcal{A}^* = (\mathcal{A}^*_1, \ldots, \mathcal{A}^*_n)$ be the allocation that minimizes the maximum envy with respect to the true values, $\max_{i,j\in[n]} v^*_i(\mathcal{A}^*_j) - v^*_i(\mathcal{A}^*_i)$. By Lemma 4, this is at most $-c'_{\mathcal{D}} \cdot m$ with probability $1 - \exp(-\Omega_{\mathcal{D}}(m))$, where $c'_{\mathcal{D}}$ is defined in the lemma. Condition on this event.

Let $\hat{\mathcal{A}} = (\hat{A}_1, \ldots, \hat{A}_n)$ be the allocation that minimizes the maximum *observed* envy, $\max_{i,j\in[n]} \hat{v}_i(A_j) - \hat{v}_i(A_i)$. Since $\mathcal{A}^*$ is a feasible allocation, β can be no larger than the observed envy gap of $\mathcal{A}^*$, which can be at most $\varepsilon \cdot m$ larger than its true gap. Thus, $\beta \leq -c'_{\mathcal{D}} \cdot m + \varepsilon \cdot m$.

We will choose $c_{\mathcal{D}} = c'_{\mathcal{D}}/4$, so $\varepsilon \leq c'_{\mathcal{D}}/4$. Plugging this in, we get that $\beta \leq -(3/4) \cdot c'_{\mathcal{D}} m$.

Now consider the following linear program that minimizes the maximum envy (with respect to observed values) over all fractional allocations.

$$\begin{aligned}
&\min \alpha \\
\alpha \geq \sum_{j=1}^m \hat{v}_{i,j}x_{i',j} - \hat{v}_{i,j}x_{i,j} &\qquad \forall i \in [n], i' \in [n], i' \neq i \\
\sum_{i=1}^n x_{i,j} = 1 &\qquad \forall j \in [m] \\
x_{i,j} \geq 0 &\qquad \forall i \in [n], j \in [m]
\end{aligned}$$

The integral allocation $\hat{\mathcal{A}}$ is a feasible solution for this LP, therefore, the value of optimal solution of this LP, x^*, has maximum envy at most β. Let $\mu_{i,i'} = \sum_{j=1}^{m} x^*_{i',j}\hat{v}_{i,j}$. Let $\tilde{\mathcal{A}} = (\tilde{A}_1, \ldots, \tilde{A}_n)$ be the integral allocation we get by allocating item j to agent i with probability $x^*_{i,j}$, and let $\mu_{i,i'} = \mathbb{E}[\hat{v}_i(\tilde{A}_{i'})]$. Note that $\mu_{i,i'} - \mu_{i,i} \leq \beta \leq -(3/4)c'_{\mathcal{D}}m$.

The expected observed value from agent i for bundle $\tilde{A}_{i'}$ is $\mathbb{E}[\hat{v}_i(\tilde{A}_{i'})] = \mu_{i,i'}$. The value $\hat{v}_i(\tilde{A}_{i'})$ is the sum of m independent random variables (one for each item), each bounded in the range $[0, \hat{v}_{i,j}] \subseteq [0, b+\varepsilon]$. Since $\varepsilon \leq c_{\mathcal{D}}$, the range is $[0, b + c_{\mathcal{D}}]$. By Hoeffding's inequality, for any $\delta > 0$:

$$\Pr[|\hat{v}_i(\tilde{A}_{i'}) - \mu_{i,i'}| \geq \delta] \leq 2\exp\left(\frac{-2\delta^2}{m(b+c_{\mathcal{D}})^2}\right).$$

Applying a union bound over all $\leq n^2$ pairs, the probability that any observed envy deviates from its mean by more than δ is

$$\Pr[\forall i, i' : |\hat{v}_i(\tilde{A}_{i'}) - \mu_{i,i'}| \leq \delta] \geq 1 - 2\exp\left(\frac{-4\delta^2 \ln(n)}{m}\right).$$

We will choose $\delta = m \cdot c'_{\mathcal{D}}/4$. This ensures that except with probability $\exp(-\Omega(m))$, $\hat{v}_i(\tilde{A}_{i'}) - \hat{v}_i(\tilde{A}_i) \leq \beta + 2\delta \leq -m \cdot c'_{\mathcal{D}}/4$.

Finally, note that $v^*_i(\tilde{A}_{i'}) - v^*_i(\tilde{A}_i)$ can deviate from the observed gap by at most $m \cdot \varepsilon \leq m \cdot c'_{\mathcal{D}}/4$. This ensures that $\tilde{A}$ is envy-free. A union bound over the two events yields the theorem. □

Lemma 4. *Let $\mathcal{D}$ be a distribution satisfying the above conditions. Assume that $v^*_j \sim \mathcal{D}$, independently, for all items j. Then, there is a constant $c'_{\mathcal{D}}$ depending only on $\mathcal{D}$ such that with probability at least $1 - \exp(-\Omega_{\mathcal{D}}(m))$, there exists an allocation $\mathcal{A}^*$ such that, for all agents $i, j \in [n]$, $v^*_i(A^*_i) \geq v^*_i(A^*_j) + c'_{\mathcal{D}}\, m$.*

The proof of Lemma 4 relies on the following theorem about the existence of strongly envy-free allocations in cake cutting. Recall, given a σ-algebra W over a set C, a function $\mu : W \to \mathbb{R}_{\geq}0$ is a *non-atomic probability measure* if (a) $\mu_i(A) \geq 0$ for every $A \in W$, (b) $\mu_i(\emptyset) = 0$, (c) given a countable collection of pairwise disjoint elements of W, $A_1, A_2, \ldots$, $\mu_i(\bigcup_j A_j) = \sum_{j=1}^{\infty} \mu_i(A_j)$, (d) if $\mu_i(A) > 0$, then for some $B \subseteq A$ with $B \in W$, $0 < \mu_i(B) < \mu_i(A)$, and (e) $\mu_i(C) = 1$.

Theorem 8 (Theorem 5.5 (b) of [13])**.** *Fix n non-atomic probability measures $\mu_1, \ldots, \mu_n$ over a set C with σ-algebra W. Suppose that no two measures are identical. Then, there exists a partition $(P_1, \ldots, P_n)$ of C such that $\mu_i(P_i) > \mu_i(P_j)$ for all $i \neq j$.*

We are now ready to prove the lemma.

Proof of Lemma 4. Fix such a distribution $\mathcal{D}$ satisfying the conditions. Let $X = (X_1, \ldots, X_n) \sim \mathcal{D}$ be a random vector.

The key to this lemma is to represent $\mathcal{D}$ as n different valuations over the unit interval $[0, 1]$ and to find a *strongly envy-free allocation* of $[0, 1]$. This allocation will correspond to a strongly envy-free randomized allocation. By taking a sufficiently large number of samples, we can convert this into a strongly envy-free integral allocation using concentration inequalities.

Recall that X_i is bounded in $[0, b]$ for some bound b.

By Lemma 4.22 of [30], there exists a measurable function $f : [0, 1] \to [0, b]$ such that $f(U)$ has distribution $\mathcal{D}$ when $U \sim \text{Unif}[0, 1]$. For each i, let f_i denote the i-th coordinate function of f, i.e., $f_i : [0, 1] \to [0, b]$. Define the following n set functions $\mu_1, \ldots, \mu_n$ on the Borel sigma-algebra $\mathcal{B}([0, 1])$:

$$\mu_i(A) = \int_A f_i \, d\lambda, \quad \text{for all } A \in \mathcal{B}([0, 1]),$$

where λ is the Lebesgue measure. Since $\mathbb{E}[|X_i|]$ is finite, each f_i is integrable, and μ_i is a measure on $[0, 1]$. It is nonnegative as f_i is nonnegative. Since f_i is the Radon-Nikodym derivative of μ_i with respect to Lebesgue measure, each μ_i is absolutely continuous and hence nonatomic. We have that $\mu_i(\emptyset) = 0$, and it satisfies countable additivity by properties of Lebesgue integration.

For each i, let $\mu_i'(A) = \mu_i(A)/\mu_i([0,1])$ be its normalized measure. Note that this is well-defined, as if $\mu_i([0,1]) = 0$, then $\mu_i(A) = 0$ for all Borel A, which implies that $f_i(U) = 0$ with probability 1. This means that $\Pr[X_i = 0 \cdot X_j] = 1$ for $j \neq i$, contradicting the assumption on $\mathcal{D}$.

The Radon-Nikodym derivative of μ_i' is $f_i'(x) = f_i(x)/\mu_i([0,1])$. If $\mu_i' = \mu_j'$, their derivatives must be equal almost everywhere (see Theorem 3.8 of [26]), i.e., $f_i' = f_j'$ almost everywhere. This implies that $f_i(x) = c \cdot f_j(x)$ for $c = \mu_i([0,1])/\mu_j([0,1])$ almost everywhere on $[0,1]$, which again implies that $\Pr[X_i = c \cdot X_j] = 1$, breaking our condition on $\mathcal{D}$.

Using Theorem 8, there is a partition of $[0,1]$ among the agents $B_1, \ldots, B_n$ such that $\mu_i'(B_i) > \mu_i'(B_j)$ for $i \neq j$. Note that this immediately implies that $\mu_i(B_i) \geq \mu_i(B_j)$ for all $i \neq j$ as well.

Define:

$$\varepsilon = \min_{i \neq j}(\mu_i(B_i) - \mu_i(B_j)),$$

which is positive since it is the minimum over a finite number of strictly positive values.

We now interpret $\mu_1, \ldots, \mu_n$ as valuation functions over $[0,1]$. We observe:

$$\mathbb{E}[f_i(U) \cdot \mathbb{1}[U \in B_i]] = \int_{B_i} f_i \, d\lambda = \mu_i(B_i),$$

and thus for any $i \neq j$:

$$\mathbb{E}[f_i(U) \cdot (\mathbb{1}[U \in B_i] - \mathbb{1}[U \in B_j])] = \mu_i(B_i) - \mu_i(B_j) \geq \varepsilon.$$

We now sample m i.i.d. draws $X^1, \ldots, X^m \sim \mathcal{D}$ and show that with high probability, we obtain a strongly envy-free allocation. We can couple these draws with uniform draws by sampling $U^1, \ldots, U^m \sim \text{Unif}[0,1]$ i.i.d., and setting

$$X^j = f(U^j).$$

We define an allocation $\mathcal{A}$ by:

$$A_i = \{k \mid U^k \in B_i\}.$$

Set $c_{\mathcal{D}}' = \varepsilon/2$, which so far depends only on the distribution $\mathcal{D}$ and not m. We will show that with probability $1 - \exp(-\Omega_{\mathcal{D}}(m))$, we have:

$$v_i(A_i) - v_i(A_j) > c_{\mathcal{D}}' \cdot m, \quad \forall i \neq j.$$

Fix $i \neq j$. Define:

$$Y_{i,j}^k = X_i^k \cdot (\mathbb{1}[k \in A_i] - \mathbb{1}[k \in A_j]).$$

Observe that $v_i(A_i) - v_i(A_j) = \sum_k Y_{i,j}^k$. We would like to show that for all $i \neq j$, $\sum_k Y_{i,j}^k \geq c_{\mathcal{D}}' \cdot m$. To do so, we bound the complement: $\Pr[\sum_k Y_{i,j}^k < c_{\mathcal{D}}' \cdot m]$. Note that $\mathbb{E}[\sum_k Y_{i,j}^k] \geq m \cdot \varepsilon = 2c_{\mathcal{D}}' \cdot m$. Thus, this is a deviation from the mean of $> c_{\mathcal{D}}' \cdot m$. Furthermore, each $Y_{i,j}^k \in [-b, b]$, $\mathbb{E}[Y_k] \geq \varepsilon$, the Y_ks are independent. We can therefore apply Hoeffding's inequality to get:

$$\Pr\left[\sum_k Y_k < c \cdot m\right] \leq \exp\left(-\frac{c^2 m}{2b^2}\right).$$

Applying a union bound over all $n(n-1)$ pairs:

$$\Pr\left[\exists i \neq j, v_i(A_i) - v_i(A_j) < c \cdot m\right] \leq n(n-1)\exp\left(-\frac{c^2 m}{2b^2}\right) \in \exp\left(-\Omega_{\mathcal{D}}(m)\right),$$

completing the proof. □

The purpose of the non-identical condition is now clearer. If two agents had the same values (up to scaling), then it is impossible that both strongly prefer their own bundles.

D Missing Proofs from Section 4.1

Theorem 3 (Upper bound for worst-case noise)**.** *For every $i \in [n]$ and $j \in [m]$, let $\hat{v}_{i,j}$ be an estimate of $v^*_{i,j} \in [0, b]$, such that $\max_{i,j} |v^*_{i,j} - \hat{v}_{i,j} - \delta_i| \leq \varepsilon$, for some possibly unknown constants $\varepsilon > 0$ and $\delta_i \geq 0$. Given $\hat{v}_{i,j}$ for each i, j, we can efficiently compute an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that, for every pair of agents $i, i' \in [n]$, we have $v^*_i(A_{i'}) - v^*_i(A_i) \leq 2\varepsilon \left\lceil \frac{m}{n} \right\rceil + b$.*

Proof. Consider the allocation produced by the Round-Robin algorithm, when executed on the estimated values $\hat{v}_{i,j}$. Recall that the Round-Robin algorithm allocates items to agents over a sequence of rounds, according to a fixed, arbitrary order. In each round, the next agent in the order picks an item that maximizes her value ($\hat{v}_{i,j}$, in our case), among the set of remaining items (ties can be broken in an arbitrary manner). The allocation $\mathcal{A}$ produced by Round-Robin satisfies envy-freeness up to one item (EF1) *with respect to the noisy values*. That is, for any pair of agents $i, i' \in [n]$ with $\hat{v}_i(A_{i'}) \neq \emptyset$, we have $\hat{v}_i(A_i) \geq \hat{v}_i(A_{i'} \setminus \{g\})$ for some $g \in A_{i'}$. Furthermore, the allocation $\mathcal{A}$ produced by Round-Robin is *balanced*, i.e., all agents get roughly equal number of items, for each agent $i \in [n]$, $|A_i|$ is either $\lfloor \frac{m}{n} \rfloor$ or $\lceil \frac{m}{n} \rceil$.

Given these facts, we will bound the envy of an agent $i \in [n]$ for agent $i' \in [n]$ with respect to the true (unknown) values as follows. If $A_{i'} = \emptyset$, then there is no envy since $v^*_i(A_{i'}) - v^*_i(A_i) = -v^*_i(A_i) \leq 0$. Otherwise, if $A_{i'} \neq \emptyset$, let $g \in A_{i'}$ be an item such that $\hat{v}_i(A_i) \geq \hat{v}_i(A_{i'} \setminus \{g\})$. Additionally, note that the inequality $|v^*_{i,j} - \hat{v}_{i,j} - \delta_i| \leq \varepsilon$ implies that (i) $v^*_i(A_i) \geq \hat{v}_i(A_i) + \delta_i|A_i| - \varepsilon|A_i|$ and (ii) $v^*_i(A_{i'} \setminus \{g\}) \leq \hat{v}_i(A_{i'} \setminus \{g\}) + \delta_i|A_{i'} \setminus \{g\}| + \varepsilon|A_{i'} \setminus \{g\}|$. Using this we get,

$$
\begin{aligned}
& v^*_i(A_{i'}) - v^*_i(A_i) \\
& \quad \leq v^*_i(A_{i'} \setminus \{g\}) - v^*_i(A_i) + b && (v^*_{i,j} \leq b \text{ for all } j \in [m]) \\
& \quad \leq \hat{v}_i(A_{i'} \setminus \{g\}) + \delta_i|A_{i'} \setminus \{g\}| + \varepsilon|A_{i'} \setminus \{g\}| - v^*_i(A_i) + b && (\text{Via } (i)) \\
& \quad \leq (\hat{v}_i(A_{i'} \setminus \{g\}) - \hat{v}_i(A_i)) + \delta_i \left(|A_{i'} \setminus \{g\}| - |A_i|\right) + \varepsilon \left(|A_i| + |A_{i'} \setminus \{g\}|\right) + b \\
& & & (\text{Via } (ii)) \\
& \quad \leq (\hat{v}_i(A_{i'} \setminus \{g\}) - \hat{v}_i(A_i)) + \varepsilon \left(|A_i| + |A_{i'} \setminus \{g\}|\right) + b \\
& & & (\delta_i \geq 0, |A_i|, |A_{i'}| \in \{\lfloor \tfrac{m}{n} \rfloor, \lceil \tfrac{m}{n} \rceil\}) \\
& \quad \leq \hat{v}_i(A_{i'} \setminus \{g\}) - \hat{v}_i(A_i) + 2\varepsilon \left\lceil \frac{m}{n} \right\rceil + b && (|A_i|, |A_{i'}| \leq \lceil \tfrac{m}{n} \rceil) \\
& \quad \leq 2\varepsilon \left\lceil \frac{m}{n} \right\rceil + b. && (\hat{v}_i(A_{i'} \setminus \{g\}) \leq \hat{v}_i(A_i))
\end{aligned}
$$

This concludes the proof. □

Theorem 4 (Lower bound for worst-case noise)**.** *For every $i \in [n]$ and $j \in [m]$, let $\hat{v}_{i,j}$ be an estimate of $v^*_{i,j}$, such that $\max_{i,j} |v^*_{i,j} - \hat{v}_{i,j}| \leq \varepsilon$, for some $\varepsilon > 0$. No deterministic algorithm that takes $\hat{v}_{i,j}$s as input, and outputs a complete allocation, can guarantee that the envy between any two agents is strictly less than $2\varepsilon \frac{m}{n}$.*

Proof. Consider any deterministic algorithm, and let $\mathcal{A} = (A_1, \ldots, A_n)$ be the allocation returned by this algorithm when $\hat{v}_{i,j} = 1/2$ for all $i \in [n]$ and $j \in [m]$. Let $p \in \arg\max_{i \in [n]} |A_i|$ and $q \in \arg\max_{i \in [n] \setminus \{p\}} |A_i|$ be the agent with the largest and second largest number of items in $\mathcal{A}$, respectively. We have that $|A_p| + |A_q| \geq \frac{2m}{n}$. We will construct the true values $v^*_{i,j}$s to maximize the envy of agent q for agent p. In particular, we set $v^*_{q,j} = 1/2 + \varepsilon$ for all $j \in A_p$ and $v^*_{q,j} = 1/2 - \varepsilon$ for all $j \in A_q$. This implies that the envy of agent q for agent p, $v^*_q(A_p) - v^*_q(A_q) = |A_p|(1/2 + \varepsilon) - |A_q|(1/2 - \varepsilon) = \varepsilon(|A_p| + |A_q|) + 1/2(|A_p| - |A_q|)$. Using the fact that $|A_p| \geq |A_q|$ and $|A_p| + |A_q| \geq \frac{2m}{n}$, we get $v^*_q(A_p) - v^*_q(A_q) \geq 2\varepsilon \frac{m}{n}$. □

E Missing Proofs from Section 4.2

Application 1 (Inference and Fair Division)**.** *Assume there are m indivisible items and n additive agents with $n < m$, where $v^*_{i,j} \in [0, 1]$ is the value of agent i for item j. Let $V^* \in [0, 1]^{n \times m}$ be the valuation matrix where $V^*_{i,j} = v^*_{i,j}$. If agents respond to BTL-based pairwise queries, then there is an algorithm that asks $\binom{m}{2} pK$ average number of pairwise-comparison queries to each agent*

$i \in [n]$, and with probability at least $1 - O(1/m^6)$, produces an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ in polynomial time that has bounded maximum envy. Formally, the obtained allocation $\mathcal{A}$ satisfies $\max_{i,j} v_i(A_j) - v_i(A_i) = O\left(\sqrt{\frac{m \log m}{n^2 pK}}\right)$, where K is the number of queries per-pair and p is the observation graph probability satisfying $p = \Omega\left(\frac{\log m}{m}\right)$.

Proof of Application 1. We begin by transforming the true valuation matrix $V^* \in [0,1]^{n \times m}$ into a preference score matrix Θ^* via a centering operation. Specifically, for each agent $i \in [n]$ and item $j \in [m]$, we define

$$\theta^*_{i,j} = b\left(v^*_{i,j} - \frac{1}{m}\sum_{l=1}^{m} v^*_{i,l}\right), \tag{6}$$

where $b > 0$ is a scaling parameter (assumed to be constant). Note that the centering in Eq. (6) ensures that $\Theta^* \mathbf{1}_m = \mathbf{0}_n$. Furthermore, as $v^*_{i,j}$s lie in $[0,1]$, we have $\theta^*_{i,j} \in (-b, b)$.

Next, for every row of Θ^*, that is each vector $\theta^{*\mathrm{T}}_i$, we invoke Lemma 5 on θ^*_i. Under the conditions specified in that theorem, each invocation returns an estimate $\tilde{\theta}_i$ satisfying

$$\|\tilde{\theta}_i - \theta^*_i\|_\infty = O\left(\sqrt{\frac{\log m}{mpK}}\right),$$

with probability at least $1 - O(1/m^7)$. Therefore, by taking $\hat{\Theta} = [\tilde{\theta}_1; \ldots; \tilde{\theta}_n]^{\mathrm{T}}$, we have

$$\max_{i,j} |\hat{\Theta}_{i,j} - \Theta^*_{i,j}| = O\left(\sqrt{\frac{\log m}{mpK}}\right) \tag{7}$$

Through a union bound, we get that Equation (7) holds with probability at least $1 - O(n/m^7)$, which is at least $1 - O(1/m^6)$ since $n < m$. Finally, we define the estimated valuation matrix as $\hat{V} := \hat{\Theta}/b$, and for each agent i, we define $\delta_i := \frac{1}{m}\sum_{j=1}^{m} v^*_{i,j} \in [0,1]$, which represents the mean valuation of agent i. Using this along with Equation (6) and (7), for any $i \in [n]$ and $j \in [m]$, we get that

$$\left|\hat{v}_{i,j} - \left(v^*_{i,j} - \frac{1}{m}\sum_{l=1}^{m} v^*_{i,j}\right)\right| = \left|\hat{v}_{i,j} - \Theta^*_{i,j}/b\right| = \frac{1}{b}\left|\hat{\Theta}_{i,j} - \Theta^*_{i,j}\right| = O\left(\sqrt{\frac{\log m}{mpK}}\right),$$

where in the last step we assume b is some constant. This implies that, for each $i \in [n]$ and $j \in [m]$, we have an estimate $\hat{v}_{i,j}$ that satisfies $|v^*_{i,j} - \hat{v}_{i,j} - \delta_i| = |\hat{v}_{i,j} - v^*_{i,j} + \delta_i| = O\left(\sqrt{\frac{\log m}{mpK}}\right)$ for the value of $\delta_i = \frac{1}{m}\sum_{l=1}^{m} v^*_{i,j} \in [0,1]$. Note that while $\hat{v}_{i,j}$ is known, δ_i is unknown. Regardless, we can apply Theorem 3 to compute an allocation $\mathcal{A} = (A_1, A_2, \ldots, A_n)$ in polynomial time that has bounded maximum envy with respect to the true values, i.e., $\max_{i,j} v^*_i(A_j) - v^*_i(A_i) = O\left(\varepsilon \cdot \frac{m}{n}\right) = O\left(\sqrt{\frac{m \log m}{n^2 pK}}\right)$. This concludes the proof. □

Application 2. *For every $i \in [n]$ and $j \in [m]$, let $z_{i,j}$s be i.i.d. draws from a non-negative MHR distribution $\mathcal{D}$ such that $\mathbb{E}[\mathcal{D}] \leq \frac{n}{m \log(nm)}$, and $\sigma_{i,j}$s be uniformly random draws from $\{-1, 1\}$. Let $\hat{v}_{i,j} = v^*_{i,j} + \varepsilon_{i,j}$, for $\varepsilon_{i,j} = \sigma_{i,j} \cdot z_{i,j}$. Given $\hat{v}_{i,j}$ and assuming $v^*_{i,j} \in [0, b]$ for each i, j, we can efficiently compute an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that, for every pair of agents $i, i' \in [n]$, we have $v^*_i(A_{i'}) - v^*_i(A_i) \leq 2\,\log(nm)\,\mathbb{E}[\mathcal{D}]\left\lceil \frac{m}{n} \right\rceil + b$, with probability at least $1 - \frac{1}{(nm)^{3/5}}$.*

Proof of Application 2. We use the following two lemmas about MHR distributions, where $\mathcal{D}_{k:n}$ denotes the distribution of the k-th lowest of n i.i.d. samples from $\mathcal{D}$.

Lemma 6 (Lemma 5.3 of [10])**.** *For any non-negative MHR distribution $\mathcal{D}$ and $m \leq n$, it holds that $\frac{\mathbb{E}[\mathcal{D}_{m:m}]}{\mathbb{E}[\mathcal{D}_{n:n}]} \geq \frac{H_m}{H_n}$, where H_k is the k-th harmonic number.*

Lemma 7 ([9]; Lemma 3)**.** *For any non-negative MHR distribution $\mathcal{D}$, and $n \geq 4$, $\Pr[\mathcal{D}_{n:n} < 2\,\mathbb{E}[\mathcal{D}_{n:n}]] \geq 1 - \frac{1}{n^{3/5}}$.*

Let $\varepsilon_{\max} = \max_{i,j} |\varepsilon_{i,j}|$ be the magnitude of the largest noise in an instance. $\varepsilon_{\max}$ follows the same distribution as $\mathcal{D}_{nm:nm}$. Therefore, letting $\mu_{\max} = \mathbb{E}[\mathcal{D}_{nm:nm}]$, by Lemma 7, $\Pr[\varepsilon_{\max} < 2\,\mu_{\max}] \geq 1 - \frac{1}{(nm)^{3/5}}$. However, by Lemma 6, $\mu_{\max} = \mathbb{E}[\mathcal{D}_{nm:nm}] \leq H_{nm}\mathbb{E}[\mathcal{D}] \leq \log(nm)\mathbb{E}[\mathcal{D}]$. Therefore,

$$\Pr[\varepsilon_{\max} < 2\,\log(nm)\,\mathbb{E}[\mathcal{D}]] \geq 1 - \frac{1}{(nm)^{3/5}}.$$

Let $\beta = 2\,\log(nm)\,\mathbb{E}[\mathcal{D}]$, and condition on the event that $\varepsilon_{\max} < \beta$. Then, applying Theorem 3 for $\delta_i = 0$ for all agents i, we can find an allocation $\mathcal{A} = (A_1, \ldots, A_n)$ such that $\max_{i,i'} v^*(A_{i'}) - v^*(A_i) \leq 2\beta \left\lceil \frac{m}{n} \right\rceil + b$. □

F Missing Proofs from Section 4.3

F.1 Missing Proofs from Section 4.3.1

We first want to establish the discrepancy bound of Theorem 6. To do this, we will show that the high-dimensional random vector w_t behaves like a subgaussian.[10] We begin by introducing relevant definitions and a useful lemma.

Definition 2 ([36])**.** *Every σ-subgaussian random vector $X \in \mathbb{R}^n$ satisfies the following bounds*

1. $\mathbb{E}[\exp(\langle X, u\rangle^2/\sigma^2)] \leq 2$ *for all* $u \in \mathcal{S}^{n-1}$ *and*[11]
2. *If* $\mathbb{E}[X] = 0$*, then* $\mathbb{E}[\exp(\langle X, \theta\rangle)] \leq \exp(C\sigma^2\|\theta\|_2^2)$ *for all* $\theta \in \mathbb{R}^n$*, where* $C > 0$ *is a universal constant, independent of the random vector.*

Lemma 8 (Hoeffding's Lemma)**.** *Let X be any random variable that lies in the range $[a, b]$. Then, for any constant $\lambda \in \mathbb{R}$, we have* $\mathbb{E}[\exp(\lambda X)] \leq \exp(\lambda\mathbb{E}[X] + \frac{\lambda^2(b-a)^2}{8})$.

We will now define an event that will play a crucial role in the analysis of Algorithm 1. Let $\mathrm{Fail_t}$ be the event that during the start of iteration t, w_t satisfies the following condition,

$$\mathrm{Fail_t} \coloneqq \mathbb{1}[\langle w_t, v_t^* + \eta_t\rangle > 4(1-p)c_t] \vee \mathbb{1}[\langle w_t, v_t^* + \eta_t\rangle < -4pc_t] \vee \mathbb{1}[\|w_t\|_\infty > c_t/\sqrt{n}] \quad (8)$$

That is, $\mathrm{Fail_t}$ is true if either $\|w_t\|_\infty$ is large or $(1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t} \notin [0, 1]$. Note that if the latter condition does not hold, i.e., $(1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t} \in [0, 1]$, then we have $r\left((1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}\right) = (1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}$. We additionally define $\mathrm{Fail_t^c}$ be the complement event of $\mathrm{Fail_t}$. The following lemma proves a concentration result for the random vector w_t, showing that it satisfies the definition of subgaussianity (Definition 2), with an extra multiplicative factor.

Lemma 9. *For all $t \in [m]$, the random vector $w_t \in \mathbb{R}^n$ satisfies the following properties:*

1. *for all $\theta \in \mathbb{R}^n$, we have* $\mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \theta\rangle) \mid \cap_{\ell=1}^{t-1}\mathrm{Fail_\ell^c}] \leq \left(1 - \frac{\delta}{m}\right)^{1-t}\exp(\sigma_t^2\|\theta\|_2^2)$ *where* $\sigma_t^2 = c_t = \frac{9n}{4C}\log\frac{5nm}{\delta} + \varepsilon nt$.
2. $\mathbb{P}[\mathrm{Fail_t^c} \mid \cap_{\ell=1}^{t-1}\mathrm{Fail_\ell^c}] \geq 1 - \frac{\delta}{m}$.

Proof. We will prove (1) and (2) property by induction on t. Since $w_1 = \mathbf{0}$, both (1) and (2) are trivially satisfied for $t = 1$. Now assuming that (1) and (2) are true for w_t, we now prove that both of these hold for w_{t+1} as well; this will complete the inductive argument.

Proving property (1)**.** For any $\theta \in \mathbb{R}^n$ consider,

$$\mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^{t}\mathrm{Fail_\ell^c}]$$

[10] In Lemma 9, we will show that w_t satisfies the definition of subgaussianity, albeit with an additional multiplicative factor.
[11] The set $\mathcal{S}^{n-1} \coloneqq \{x \in \mathbb{R}^n \mid \|x\|_2 = 1\}$ denotes the unit sphere in n dimension.

$$= \mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_t + s_t v_t^*, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \qquad (w_{t+1} = w_t + s_t v_t^* \text{ given } \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}})$$

$$= \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}\left[\exp(\langle w_t, \theta\rangle) \cdot \mathbb{E}_{s_t}[\exp(s_t\langle v_t^*, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}\right] \tag{9}$$

To proceed further, we will now invoke Hoeffding's lemma (Lemma 8) by setting $X = s_t$, $\lambda = \langle v_t^*, \theta\rangle$, $b = p$, and $a = p - 1$. This gives us

$$\begin{aligned}\mathbb{E}_{s_t}[\exp(s_t\langle v_t^*, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] &\leq \exp\left(\langle v_t^*, \theta\rangle \mathbb{E}[s_t] + \frac{\langle v_t^*, \theta\rangle^2 (p - (p-1))^2}{8}\right) \\ &= \exp\left(\frac{-\langle v_t^*, \theta\rangle\langle w_t, v_t^* + \eta_t\rangle}{4c_t} + \frac{\langle v_t^*, \theta\rangle^2}{8}\right),\end{aligned} \tag{10}$$

where the final equality uses the fact that given the event $\mathrm{Fail}_{\mathrm{t}}^{\mathrm{c}}$, we have $r\left((1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}\right) = (1-p) - \frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}$, and therefore, $E[s_t] = -\frac{\langle w_t, v_t^* + \eta_t\rangle}{4c_t}$ (Line 1). Substituting Equation (10) in Equation (9) gives us,

$$\begin{aligned}&\mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \\ &\leq \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}\left[\exp(\langle w_t, \theta\rangle) \cdot \exp\left(\frac{-\langle v_t^*, \theta\rangle\langle w_t, v_t^* + \eta_t\rangle}{4c_t} + \frac{\langle v_t^*, \theta\rangle^2}{8}\right) \Big| \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}\right] \\ &= \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}\left[\exp\left(\langle w_t, \theta\rangle - \frac{\langle v_t^*, \theta\rangle\langle w_t, v_t^* + \eta_t\rangle}{4c_t}\right) \Big| \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}\right] \exp\left(\frac{\langle v_t^*, \theta\rangle^2}{8}\right) \\ &= \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}\left[\exp\left(\left\langle w_t, \theta - \frac{\langle v_t^*, \theta\rangle(v_t^* + \eta_t)}{4c_t}\right\rangle\right) \Big| \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}\right] \exp\left(\frac{\langle v_t^*, \theta\rangle^2}{8}\right)\end{aligned} \tag{11}$$

Note that our inductive hypothesis, in particular property (1) for w_t does not apply to Equation (11) since the event being conditioned on is $\cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}$ and not $\cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}$. Nevertheless, using the inductive hypothesis, we can prove the following useful claim.

Claim 1. *For all $\beta \in \mathbb{R}^n$, we have* $\mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp(\sigma_t^2 \|\beta\|_2^2)$.

Proof. Using the law of total probability we get,

$$\begin{aligned}\mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] &= \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \cdot \mathbb{P}[\mathrm{Fail}_{\mathrm{t}}^{\mathrm{c}} \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \\ &\quad + \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \mathrm{Fail}_{\mathrm{t}}, \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \cdot \mathbb{P}[\mathrm{Fail}_{\mathrm{t}} \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \\ &\geq \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \cdot \mathbb{P}[\mathrm{Fail}_{\mathrm{t}}^{\mathrm{c}} \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \\ &\geq \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \left(1 - \frac{\delta}{m}\right),\end{aligned}$$

where the final inequality follows from our induction hypothesis, i.e.,

$$\mathbb{P}[\mathrm{Fail}_{\mathrm{t}}^{\mathrm{c}} \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \geq 1 - \frac{\delta}{m}.$$

Hence, we get

$$\mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \leq \left(1 - \frac{\delta}{m}\right)^{-1} \mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}].$$

Substituting the bound $\mathop{\mathbb{E}}_{s_1,\ldots,s_{t-1}}[\exp(\langle w_t, \beta\rangle) \mid \cap_{\ell=1}^{t-1} \mathrm{Fail}_\ell^{\mathrm{c}}] \leq \left(1 - \frac{\delta}{m}\right)^{1-t} \exp(\sigma_t^2 \|\beta\|_2^2)$, from the induction hypothesis, completes the proof. □

By setting $\beta = \theta - \frac{\langle v_t^*, \theta\rangle (v_t^* + \eta_t)}{4c_t}$ in Claim 1 and substituting in Equation (11) we get

$$
\begin{aligned}
&\left(1 - \frac{\delta}{m}\right)^t \mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \\
&\leq \exp\left(\sigma_t^2 \cdot \left\|\theta - \frac{\langle v_t^*, \theta\rangle (v_t^* + \eta_t)}{4c_t}\right\|_2^2\right) \exp\left(\frac{\langle v_t^*, \theta\rangle^2}{8}\right) \\
&= \exp\left(\sigma_t^2 \cdot \left(\|\theta\|_2^2 + \frac{\langle v_t^*, \theta\rangle^2}{16c_t^2}\|v_t^* + \eta_t\|_2^2 - \frac{\langle v_t^*, \theta\rangle(\langle v_t^*, \theta\rangle + \langle \eta_t, \theta\rangle)}{2c_t}\right)\right) \exp\left(\frac{\langle v_t^*, \theta\rangle^2}{8}\right) \\
&= \exp(\sigma_t^2\|\theta\|_2^2)\, \exp\left(\langle v_t^*, \theta\rangle^2 \left(\frac{\sigma_t^2}{16c_t^2}(\|v_t^*\|_2^2 + \|\eta_t\|_2^2 + 2\langle v_t^*, \eta_t\rangle) - \frac{\sigma_t^2}{2c_t} + \frac{1}{8}\right)\right) \\
&\quad \cdot \exp\left(\frac{-\sigma_t^2 \langle v_t^*, \theta\rangle\langle \eta_t, \theta\rangle}{2c_t}\right).
\end{aligned}
$$

To simplify the above expression, we will use the fact that $\langle a, b\rangle \leq \|a\|_2\|b\|_2$, $\|v_t^*\|_2 \leq \sqrt{n}$, $\|\eta_2\|_2 \leq \varepsilon\sqrt{n} \leq \sqrt{n}$, along with $\sigma_t^2 = c_t$. Substituting these gives us,

$$
\begin{aligned}
&\mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \\
&\leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp(\sigma_t^2\|\theta\|_2^2) \cdot \exp\left(\langle v_t^*, \theta\rangle^2 \left(\frac{4n}{16c_t} - \frac{1}{2} + \frac{1}{8}\right)\right) \cdot \exp\left(\frac{-\langle v_t^*, \theta\rangle\langle \eta_t, \theta\rangle}{2}\right).
\end{aligned}
$$

Note that for a small value of δ, the term $\frac{4n}{16c_t} = 1/(\frac{9}{C}\log\frac{5nm}{\delta} + 4\varepsilon t)$ is negligible in comparison to the other terms. Hence, $\frac{4n}{16c_t} - \frac{1}{2} + \frac{1}{8} < 0$ and we get

$$
\begin{aligned}
\mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] &\leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp(\sigma_t^2\|\theta\|_2^2) \exp\left(\frac{-\langle v_t^*, \theta\rangle\langle \eta_t, \theta\rangle}{2}\right) \\
&\leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp(\sigma_t^2\|\theta\|_2^2) \exp\left(\frac{\varepsilon n\|\theta\|_2^2}{2}\right) \\
&\leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp\left((\sigma_t^2 + \varepsilon n)\|\theta\|_2^2\right) \\
&= \left(1 - \frac{\delta}{m}\right)^{-t} \exp\left(\sigma_{t+1}^2\|\theta\|_2^2\right),
\end{aligned}
$$

where the last equality follows from the fact that $\sigma_{t+1}^2 = \frac{9n}{4C}\log\frac{5nm}{\delta} + \varepsilon n(t+1)$. This establishes property (1) for w_{t+1}.

Under the assumption that $\delta \leq 1/2$, an immediate consequence of property (1), which we will use in proving property (2) is that,

$$
\begin{aligned}
\mathop{\mathbb{E}}_{s_1,\ldots,s_t}[\exp(\langle w_{t+1}, \theta\rangle) \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] &\leq \left(1 - \frac{\delta}{m}\right)^{-t} \exp\left(\sigma_{t+1}^2\|\theta\|_2^2\right) \\
&\leq \left(1 - \frac{\delta t}{m}\right)^{-1} \exp\left(\sigma_{t+1}^2\|\theta\|_2^2\right) \\
&\leq (1-\delta)^{-1} \exp\left(\sigma_{t+1}^2\|\theta\|_2^2\right) && (t \leq m) \\
&\leq 2\exp\left(\sigma_{t+1}^2\|\theta\|_2^2\right) && (12)
\end{aligned}
$$

where the second inequality follows from the fact that $(1+x)^a \geq 1 + xa$ if $x \geq -1, a \geq 1$, and final inequality follows from $\delta \leq 1/2$.

Proving property (2)**.** We will show that $\Pr[\mathrm{Fail}_{\mathrm{t+1}} \mid \cap_{\ell=1}^t \mathrm{Fail}_\ell^{\mathrm{c}}] \leq \delta/m$, which will directly prove property (2) for iteration $t+1$.

Note that Fail_{t+1} occurs only if in iteration $t+1$, $\langle w_{t+1}, v^*_{t+1}+\eta_{t+1}\rangle > 4(1-p)c_{t+1}$ or $\langle w_{t+1}, v^*_{t+1}+\eta_{t+1}\rangle < -4pc_{t+1}$ or $\|w_{t+1}\|_\infty > c_{t+1}/\sqrt{n}$ (Equation (8)). Since $p \in [1/2, 2/3]$, we have $4p > 4(1-p)$. We will utilize this fact to simplify as follows,

$$
\begin{aligned}
&\Pr[\text{Fail}_{t+1} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&= \Pr[\langle w_{t+1}, v^*_{t+1} + \eta_{t+1}\rangle > 4(1-p)c_{t+1} \\
&\qquad \text{or } \langle w_{t+1}, v^*_{t+1} + \eta_{t+1}\rangle < -4pc_{t+1} \text{ or } \|w_{t+1}\|_\infty > c_{t+1}/\sqrt{n} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&\le \Pr[|\langle w_{t+1}, v^*_{t+1} + \eta_{t+1}\rangle| > 4(1-p)c_{t+1} \text{ or } \|w_{t+1}\|_\infty > c_{t+1}/\sqrt{n} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (4p > 4(1-p)) \\
&= \Pr[|\langle w_{t+1}, (v^*_{t+1} + \eta_{t+1})/(4(1-p)\sqrt{n})\rangle| > c_{t+1}/\sqrt{n} \text{ or } \|w_{t+1}\|_\infty > c_{t+1}/\sqrt{n} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&\le \sum_{u\in S} \Pr[|\langle w_{t+1}, u\rangle| > c_{t+1}/\sqrt{n} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c],
\end{aligned}
$$

where set $S := \{(v^*_{t+1} + \eta_{t+1})/(4(1-p)\sqrt{n}), e_1, e_2, \ldots, e_n\}$ and $e_1, \ldots, e_n$ are the basis unit vectors along the axes. Note that $\forall u \in S$, we have $\|u\|_2 \le 3/2$: for all $i \in [n]$, $\|e_i\|_2 = 1$ and $\|(v^*_{t+1} + \eta_{t+1})/(4(1-p)\sqrt{n})\|_2 \le 2\sqrt{n}/(4(1-p)\sqrt{n}) \le 2/(4(1-2/3)) = 3/2$, where the last inequality uses the fact that $p \le 2/3$. We proceed to further evaluate the above probability expression,

$$
\begin{aligned}
&\Pr[\text{Fail}_{t+1} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&\le \sum_{u\in S} \Pr[|\langle w_{t+1}, u\rangle| > c_{t+1}/\sqrt{n} \mid \cap_{\ell=1}^{t}\text{Fail}_\ell^c] \\
&\le \sum_{u\in S} \Pr\left[\langle w_{t+1}, u/\|u\|_2\rangle^2 > \frac{c_{t+1}^2}{(3/2)^2 n} \,\middle|\, \cap_{\ell=1}^{t}\text{Fail}_\ell^c\right] && (\|u\|_2 \le 3/2, \forall u \in S) \\
&= \sum_{u\in S} \Pr\left[\frac{\langle w_{t+1}, u/\|u\|_2\rangle^2}{\sigma_{t+1}^2/C} > \frac{c_{t+1}C}{(3/2)^2 n} \,\middle|\, \cap_{\ell=1}^{t}\text{Fail}_\ell^c\right] && (\sigma_{t+1}^2 = c_{t+1}, \text{for } C < 1/8) \\
&= \sum_{u\in S} \Pr\left[\exp\left(\frac{\langle w_{t+1}, u/\|u\|_2\rangle^2}{\sigma_{t+1}^2/C}\right) > \exp\left(\frac{c_{t+1}C}{(3/2)^2 n}\right) \,\middle|\, \cap_{\ell=1}^{t}\text{Fail}_\ell^c\right] \\
&\le \sum_{u\in S} \mathbb{E}\left[\exp\left(\frac{\langle w_{t+1}, u/\|u\|_2\rangle^2}{\sigma_{t+1}^2/C}\right) \,\middle|\, \cap_{\ell=1}^{t}\text{Fail}_\ell^c\right] \Big/ \exp\left(\frac{c_{t+1}C}{(3/2)^2 n}\right) && \text{(Markov's inequality)}
\end{aligned}
$$

Now, we will make use of the following lemma.

Lemma 10. *For a random vector $w \in \mathbb{R}^n$ if there exists a $\lambda > 0$ such that for all $\theta \in \mathbb{R}^n$, $\mathbb{E}[\exp(\langle w, \theta\rangle)] \le 2\exp\left(C\lambda^2\|\theta\|_2^2\right)$, where $C < 1/8$ is a fixed constant, then for all $u \in \mathcal{S}^{n-1}$, we have $\mathbb{E}[\exp\left(\langle w, u\rangle^2/\lambda^2\right)] \le 3$.*

Proof. Using the Taylor series expansion we have,

$$
\mathbb{E}[\exp\left(\langle w, u\rangle^2/\lambda^2\right)] = 1 + \sum_{i=1}^{\infty} \frac{\mathbb{E}[\langle w, u\rangle^{2i}]}{i! \cdot \lambda^{2i}} \tag{13}
$$

To bound the series above, we will bound the moments $\mathbb{E}[\langle w, u\rangle^{2i}]$ for all even $i \ge 1$. Towards this note that,

$$
\begin{aligned}
\mathbb{E}[\langle w, u\rangle^{2i}] &= \int_0^\infty \mathbb{P}[\langle w, u\rangle^{2i} > x]\, dx \\
&= \int_0^\infty \mathbb{P}[\exp\left(\langle w, \alpha u\rangle\right) > \exp(\alpha x^{1/2i})]\, dx && \text{(for any } \alpha > 0) \\
&\le \int_0^\infty \frac{\mathbb{E}[\exp\left(\langle w, \alpha u\rangle\right)]}{\exp(\alpha x^{1/2i})}\, dx && \text{(Markov's inequality)}
\end{aligned}
$$

$$
\begin{aligned}
&\leq 2\int_0^\infty \frac{\exp\left(C\lambda^2\alpha^2\right)}{\exp(\alpha x^{1/2i})}\,dx && (\mathbb{E}[\exp\left(\langle w, \alpha u\rangle\right)] \leq 2\exp\left(C\lambda^2\alpha^2\right)) \\
&= 2\int_0^\infty \exp\left(-\frac{x^{1/i}}{4C\lambda^2}\right)\,dx && (\text{setting } \alpha = x^{1/2i}/(2C\lambda^2)) \\
&= 2\int_0^\infty (4C\lambda^2)^i \cdot i \cdot y^{i-1}\exp\left(-y\right)\,dy && (\text{substituting } y = \tfrac{x^{1/i}}{4C\lambda^2}) \\
&= 2i(4C\lambda^2)^i \int_0^\infty y^{i-1}\exp\left(-y\right)\,dy \\
&= 2i(4C\lambda^2)^i \cdot \Gamma(i) && (14)
\end{aligned}
$$

where the final equality follows from the definition of the well-known gamma function $\Gamma(z) := \int_0^\infty x^{z-1}e^{-z}\,dz$. Substituting the bound in Equation (14) into Equation (13), we get

$$
\begin{aligned}
\mathbb{E}[\exp\left(\langle w, u\rangle^2/\lambda^2\right)] &= 1 + \sum_{i=1}^\infty \frac{\mathbb{E}[\langle w, u\rangle^{2i}]}{i!\cdot\lambda^{2i}} \\
&\leq 1 + \sum_{i=1}^\infty \frac{2i(4C\lambda^2)^i\cdot\Gamma(i)}{i!\cdot\lambda^{2i}} \\
&\leq 1 + 2\sum_{i=1}^\infty (4C)^i && (\Gamma(i) = (i-1)!) \\
&\leq 1 + 2\sum_{i=1}^\infty 2^{-i} = 3 && (C < 1/8)
\end{aligned}
$$

This completes the proof. □

From Equation (12), we know that the random vector w_{t+1} satisfies the condition in Lemma 10 with $\lambda^2 = \sigma_{t+1}^2/C$. Hence, we get,

$$
\mathbb{E}\left[\exp\left(\frac{\langle w_{t+1}, u/\|u\|_2\rangle^2}{\sigma_{t+1}^2/C}\right) \,\Big|\, \cap_{\ell=1}^t \text{Fail}_\ell^{\text{c}}\right] \leq 3.
$$

Using the above bound, and plugging in $c_{t+1} = \frac{9n}{4C}\log\frac{5nm}{\delta} + \varepsilon n(t+1)$ in the running chain of inequalities,

$$
\begin{aligned}
\Pr[\text{Fail}_{\text{t+1}} \mid \cap_{\ell=1}^t \text{Fail}_\ell^{\text{c}}] &\leq \sum_{u\in S} 3\Big/\exp\left(\log\frac{5nm}{\delta} + \frac{4C\varepsilon(t+1)}{9}\right) \\
&= (n+1)\cdot\frac{3\delta}{5nm\exp(4C(t+1)\varepsilon/9)} && (|S| = n+1) \\
&\leq \frac{\delta}{m} && (4C(t+1)\varepsilon/9) > 0, n \geq 2)
\end{aligned}
$$

This completes the proof. □

Lemma 9 directly leads to our main theorem about vector balancing.

Theorem 6. *During the execution of Algorithm 1, the random vector $w_t = \sum_{i=1}^{t-1} s_i v_i^*$ satisfies $\|w_t\|_\infty \leq \frac{9\sqrt{n}}{4C}\log\frac{5nm}{\delta} + \varepsilon t\sqrt{n}$, for all $t \in [m]$ with probability at least $1-\delta$ for a universal constant $C > 0$.*

Proof. We will prove that with probability at least $1-\delta$, the event $\cup_{\ell=1}^m \text{Fail}_\text{t}$ does not occur during the execution of Algorithm 1, i.e., Equation (8) is never satisfied. The theorem then follows, since Equation (8) not being satisfied means that the inequality $\|w_t\|_\infty \leq c_t/\sqrt{n} = \frac{9\sqrt{n}}{4C}\log\frac{5nm}{\delta} + \varepsilon t\sqrt{n}$ holds for all t.

Using chain rule of probability we have $\Pr[\cap_{t=1}^m \text{Fail}_\text{t}^\text{c}] = \Pi_{t=1}^m \Pr[\text{Fail}_\text{t}^\text{c} \mid \cap_{\ell<t}\text{Fail}_\ell^\text{c}]$. From Lemma 9, we know that $\Pr[\text{Fail}_\text{t}^\text{c} \mid \cap_{\ell<t}\text{Fail}_\ell^\text{c}] \geq 1-\delta/m$. Hence, we get $\Pr[\cap_{t=1}^m \text{Fail}_\text{t}^\text{c}] = \Pi_{t=1}^m \Pr[\text{Fail}_\text{t}^\text{c} \mid \cap_{\ell<t}\text{Fail}_\ell^\text{c}] \geq (1-\delta/m)^m \geq 1-\delta$. This completes the proof. □

F.2 Missing Proofs from Section 4.3.2

Theorem 7. *Let $k \geq 2$ be an integer and $v_1^* + \eta_1, v_2^* + \eta_2, \ldots, v_m^* + \eta_m \in \mathbb{R}^n$ be a sequence of vectors that arrive sequentially, where $\|v_i^*\|_\infty \leq 1, \|\eta_i\|_\infty \leq \varepsilon \leq 1$ for all $i \in [m]$. When each vector $v_i^* + \eta_i$ arrives, we need to assign it a color from $[k]$ after which the vector v_i^* is revealed to us. In this model, there is an efficient algorithm that achieves a multicolored discrepancy bound of $\max_{i,j\in[k]} \| \sum_{v\in S_i} v - \sum_{v\in S_j} v\|_\infty \leq \frac{27\sqrt{n}}{2C} \log \frac{5nmk}{\delta} + 6\varepsilon m\sqrt{n}$, with probability at least $1-\delta$, where S_a is the set of vectors that are assigned the color $a \in [k]$ after all the vectors have arrived.*

Proof. Now we will show that this algorithm described in Section 4.3.2, achieves a multicolored discrepancy bound of $\max_{i,j\in[k]} \| \sum_{v\in S_i} v - \sum_{v\in S_j} v\|_\infty \leq \frac{27\sqrt{n}}{2C} \log \frac{4nmk}{\delta} + 6\varepsilon t\sqrt{n}$ with probability at least $1-\delta$. Here, for any node u, S_u is the set of vectors that pass through node u during the execution of the multicolored discrepancy algorithm. To assist with the analysis, for each node u, we define π_u to be the fraction of total leaves that are in the subtree rooted at u, i.e., the number of leaves in the subtree of u divided by k. We now state a key claim.

Claim 2. *With probability at least $1-\delta$, for all nodes u of $\mathcal{T}$, we have $\| \sum_{v\in S_u} v - \pi_u \sum_{v\in S_r} v\|_\infty \leq 3(\frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n})$*

Before proving this claim, we will show that it suffices to establish the desired multicolored discrepancy bound. Given Claim 2, for all leaves $i, j \in [k]$, we will have

$$\begin{aligned}
\Big\| \sum_{v\in S_i} v - \sum_{v\in S_j} v \Big\|_\infty &= \Big\| \Big(\sum_{v\in S_i} v - \frac{1}{k} \sum_{v\in S_r} v \Big) - \Big(\sum_{v\in S_j} v - \frac{1}{k} \sum_{v\in S_r} v \Big) \Big\|_\infty \\
&\leq \Big\| \sum_{v\in S_i} v - \pi_i \sum_{v\in S_r} v \Big\|_\infty + \Big\| \sum_{v\in S_j} v - \pi_j \sum_{v\in S_r} v \Big\|_\infty \qquad (\pi_i = \pi_j = 1/k) \\
&\overset{\text{(Claim 2)}}{\leq} 6 \cdot \left(\frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n} \right) = \frac{27\sqrt{n}}{2C} \log \frac{4nmk}{\delta} + 6\varepsilon m\sqrt{n}.
\end{aligned}$$

The above bound holds for all $i, j \in [k]$ with probability at least $1-\delta$, giving us the desired multicolored discrepancy bound. As the final step, we prove Claim 2.

Proof of Claim 2. From Theorem 6, we know that for each internal node u in $\mathcal{T}$, $\|w_m^u\|_\infty \leq \frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon t\sqrt{n}$ with probability at least $1 - \delta/k$. Since $\mathcal{T}$ has exactly k leaves, it has $k-1$ internal nodes. Therefore, through a union bound, we get that with probability at least $1-\delta$, all internal nodes u will have $\|w_m^u\|_\infty \leq \frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n}$. Additionally, note that if u_ℓ and u_r are respectively the left and right children of u, then we have $\|w_m^u\|_\infty = \|p_u \sum_{v\in S_{u_r}} v - (1-p_u) \sum_{v\in S_{u_\ell}} v\|_\infty \leq \frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n}$.

We will prove the claim by induction, starting from the root node and going down the tree $\mathcal{T}$. For the base case note that for the root node r, we have $\| \sum_{v\in S_r} v - \pi_u \sum_{v\in S_r} v\|_\infty = \| \sum_{v\in S_r} v - 1 \cdot \sum_{v\in S_r} v\|_\infty = 0$, i.e., the bound is trivially satisfied. Now, assuming that this bound is satisfied for an internal node u, we will show that it will be satisfied for both its left child u_ℓ and right child u_r. We will prove the argument only for the left child u_ℓ; the argument for the right child u_r will follow similarly.

As per our induction hypothesis, we have $\| \sum_{v\in S_u} v - \pi_u \sum_{v\in S_r} v\|_\infty \leq 3(\frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n})$. Additionally, the copy of Algorithm 1 running at a, guarantees us that $\|w_m^u\|_\infty = \|(1-p_u) \sum_{v\in S_{u_\ell}} v - p_u \sum_{v\in S_{u_r}} v\|_\infty \leq \frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m\sqrt{n}$; this follows from Theorem 6. Using this we can bound,

$$\begin{aligned}
&\| \sum_{v\in S_{u_\ell}} v - \pi_{u_\ell} \sum_{v\in S_{u_r}} v\|_\infty \\
&= \|(p_u + 1 - p_u) \sum_{v\in S_{u_\ell}} v - \pi_{u_\ell} \sum_{v\in S_{u_r}} v\|_\infty
\end{aligned}$$

$$
\begin{aligned}
&= \|(p_u + 1 - p_u) \sum_{v \in S_{u_\ell}} v - p_u \pi_u \sum_{v \in S_r} v\|_\infty && (\pi_{u_\ell} = p_u \pi_u) \\
&= \|p_u(\sum_{v \in S_u} v - \sum_{v \in S_{u_r}} v) + (1 - p_u) \sum_{v \in S_{u_\ell}} v - p_u \pi_u \sum_{v \in S_r} v\|_\infty \\
& && (\textstyle\sum_{v \in S_\ell} v = \sum_{v \in S_a} v - \sum_{v \in S_r} v) \\
&= \|p_u(\sum_{v \in S_u} v - \pi_u \sum_{v \in S_r} v) + (1 - p_u) \sum_{v \in S_{u_\ell}} v - p_u \sum_{v \in S_{u_r}} v\|_\infty \\
&\leq p_u \| \sum_{v \in S_u} v - \pi_u \sum_{v \in S_r} v\|_\infty + \|(1 - p_u) \sum_{v \in S_{u_\ell}} v - p_u \sum_{v \in S_{u_r}} v\|_\infty \\
&\leq \frac{2}{3} \| \sum_{v \in S_u} v - \pi_u \sum_{v \in S_r} v\|_\infty + \|(1 - p_u) \sum_{v \in S_{u_\ell}} v - p_u \sum_{v \in S_{u_r}} v\|_\infty && (p_a \in [\tfrac{1}{2}, \tfrac{2}{3}]) \\
&\leq \frac{2}{3} \cdot 3(\frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m \sqrt{n}) + \frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta} + \varepsilon m \sqrt{n} \\
&= 3(\frac{9\sqrt{n}}{4C} \log \frac{4nmk}{\delta}).
\end{aligned}
$$

This completes the induction step and therefore the proof. □

□

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